

Advanced Corporate Finance

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Choice of the optimal capital structure and dividend policy

3.1 The mix of sources of capital and cost of financing sources

3.2 Empirical determinants in the choice of capital structure

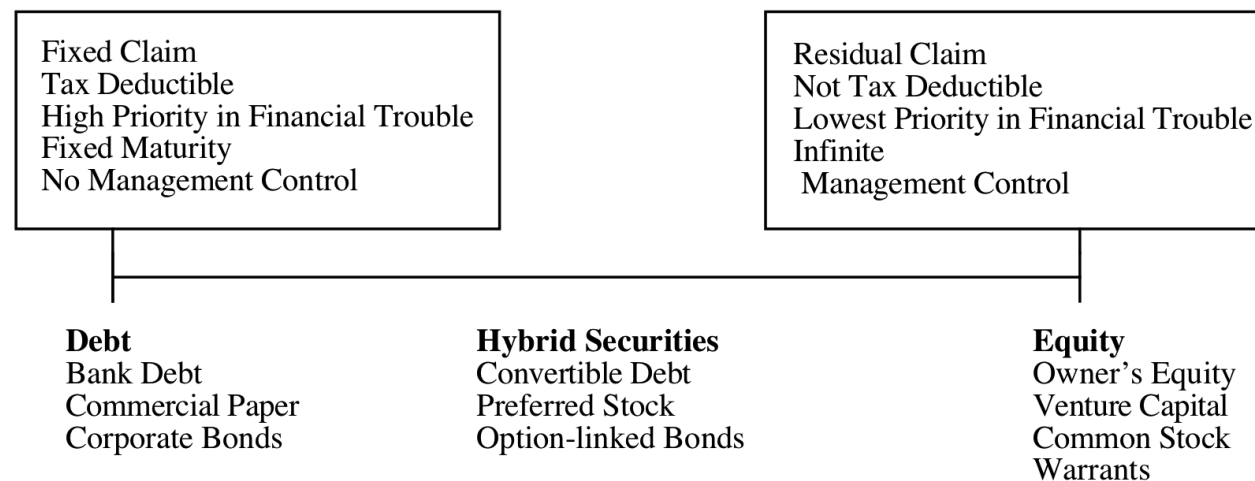
3.3 Theories of the optimal dividend policy; empirical
determinants of the optimal dividend policy

3.4 Financial structure and valuation in complete and incomplete
markets

Optimal Capital Structure

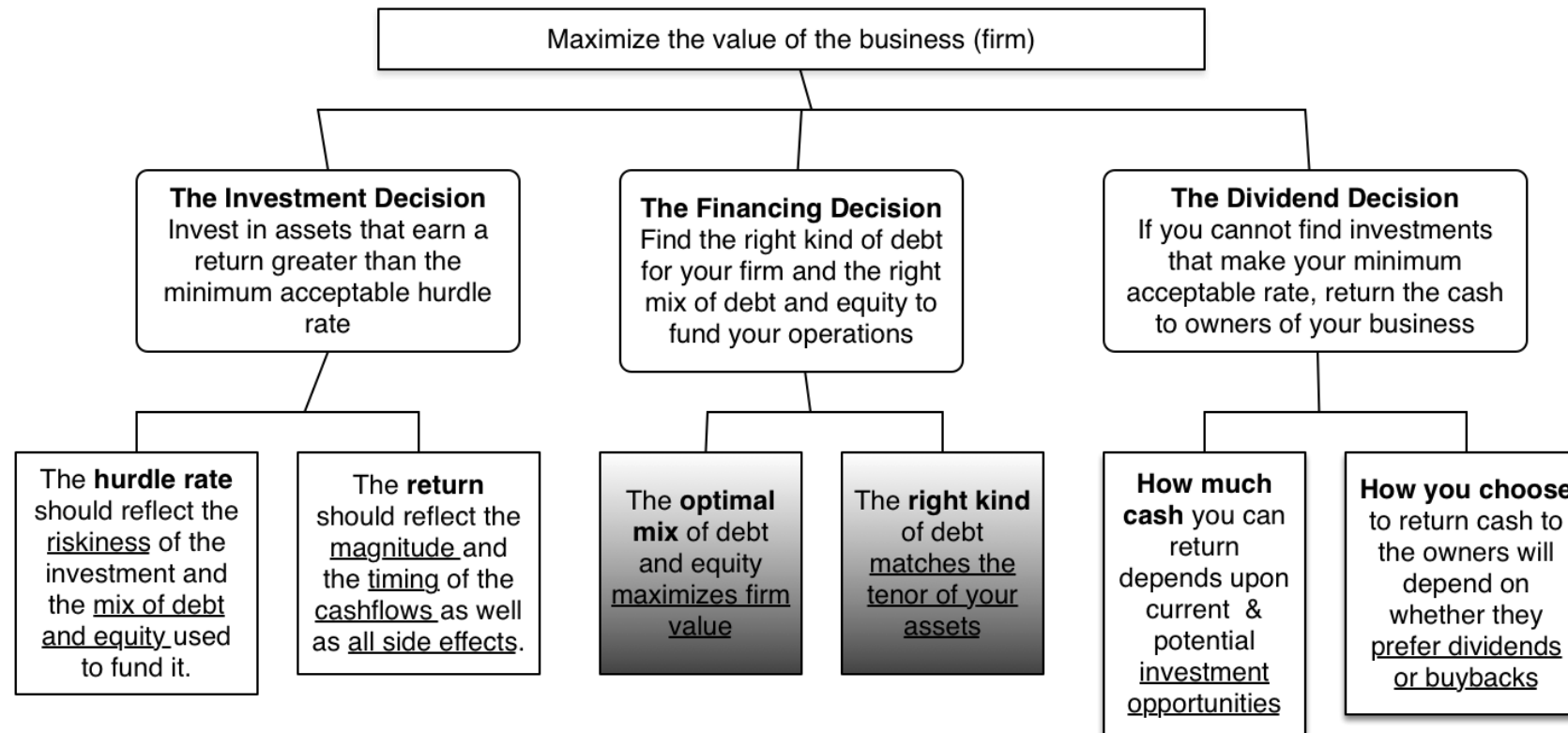
- There are only two ways a company can "create" money.
 - The first is debt. The essence of debt is that we promise to make future payments (interest and capital amortization). If we fail these payments, we lose control of the business.
 - The other is by equity. With CP, we get the cash flows left over after we make debt payments.

Figure 7.1: Debt versus Equity



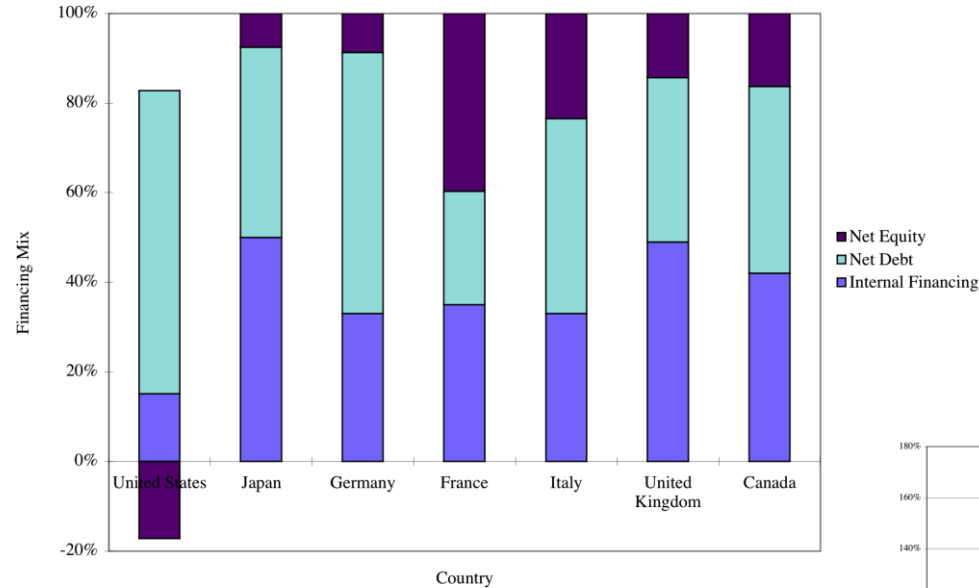
Optimal Capital Structure

- Maximum choice goal....



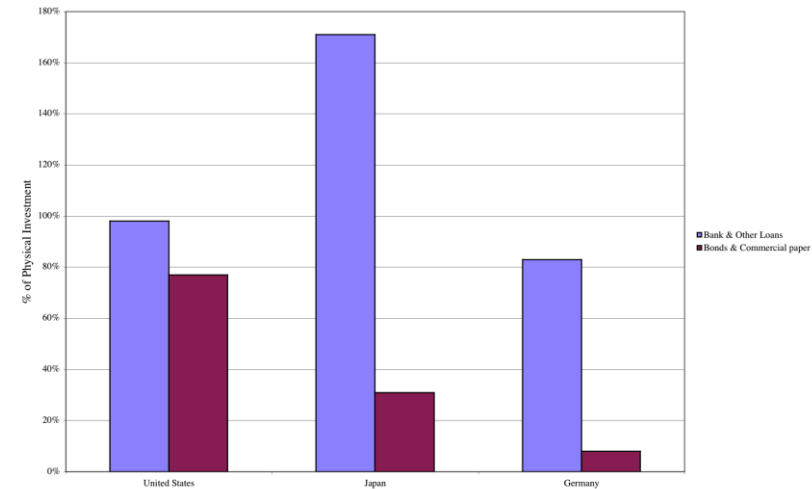
Optimal Capital Structure

Figure 7.4: Financing Patterns for G-7 Countries – 1984-91

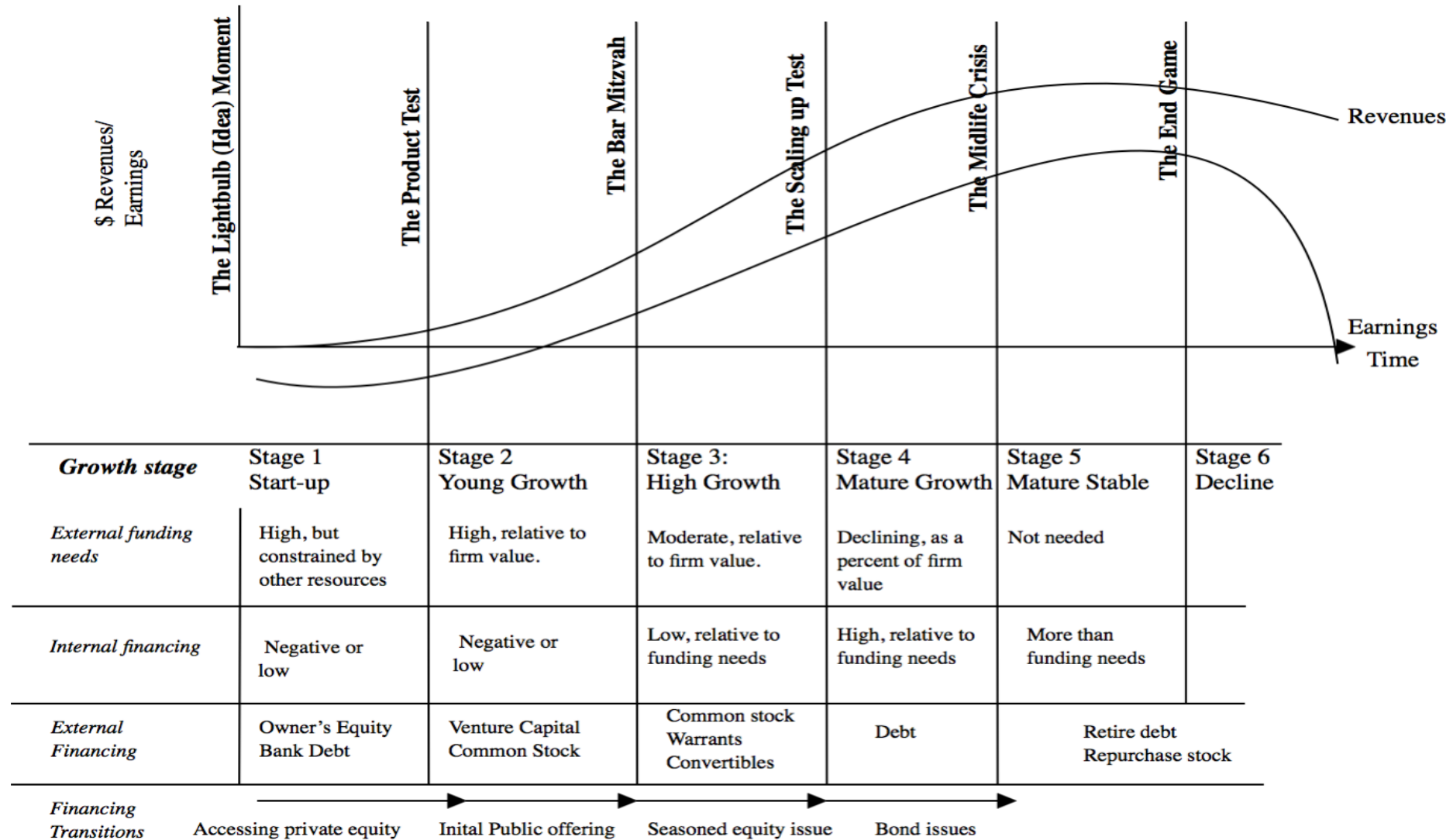


- A greater dependence for bank loans outside USA...

Figure 7.5: Bonds versus Bank Loans - 1990-96



Optimal Capital Structure



Optimal Capital Structure

- The transitions we see in companies - from privately held companies, to venture capital, from private to public and subsequent seasonal offerings, are all driven primarily by the need for capital.
- In each transition, however, there are costs incurred by existing owners:
 - When venture capitalists enter the company, they will require their fair share and more of the company's property to make capital available.
 - When a company decides to become public, it has to change the greater access to the capital markets against increasing disclosure requirements (which emanate from being publicly listed), loss of control and the capital outlay transaction costs.
 - When making seasonal offers, companies have to consider issuance costs while managing their relationships with financial analysts and ratings

Optimal Capital Structure

- The simplest way to measure how much debt and CP a company uses at present is by looking at its proportion of debt in the total funded. That is, we calculate the ratio of the degree of indebtedness
- $GE = D / A = D / (D + E)$ where D = debt, A = Assets; E = Equity
- The debt includes the associated interest, the short and the long term. It should also include other obligations reflecting debt criteria: pre-contractually established payments that have to be made regardless of the current financing situation.
- Equity can be defined in accounting terms (the book value of equity) or Market values (based on current market prices). The ratios thus obtained can be very disparate.

Optimal Capital Structure

- Does a company in deciding whether to obtain financing choose an optimal mix of debt and equity?
 - If so, what is the tradeoff that lets us determine this optimal mix?
 - What are the benefits of issuing debt rather than CP?
 - What are the costs of using debt instead of CP?
 - If not, why not?

Optimal Capital Structure

- At first glance the benefits of debt seem obvious ... the cost of debt is less than the cost of CP (Equity).
- But this benefit is illusory, because debt is cheaper than CP for a simple reason. The lender is the first to receive cash flows and has a pre-defined amount of what they receive. The CP investor is the last of the line and thus has to demand a higher rate of return than the lender.
- When borrowing money at a lower rate we are not making a business more valuable, but only to move the risk.

Optimal Capital Structure

- Benefits of debt
 - Tax Benefits: The tax code favors debt, with interest being tax deductible costs as opposed to CP.
 - Requires disciplined management: When managers are not rational in the choice of projects to finance, borrowing money can make them more disciplined.
- Cost of debt
 - Bankruptcy costs: Borrowing money will increase the likelihood and cost of bankruptcy
 - Agency Costs: What is good for stockholders is not always good for lenders and this creates frictions and costs.
 - Loss of future flexibility: Using debt capacity today means that in the future you may not be able to do so.

Optimal Capital Structure

- When we borrow money, we can deduct the interest expenses from the taxable income which reduces the fees. By using CP, we can not deduct CP payments, such as dividends, from taxable income.
- The benefit of euro taxed on interest paid in a given year is a function of the tax rate and interest:
- Annual tax benefits = Tax rate * Interest paid
- Proposition 1: Ceteris paribus, the higher the marginal tax rate, the greater the debt that will be in the capital structure.

Optimal Capital Structure

- The expected cost of bankruptcy is a function of two variables:
 - The probability of bankruptcy depends on how uncertain we are in relation to future cash flows
 - The cost of going bankrupt
 - Direct costs: Legal and other sunk costs
 - Indirect costs: costs that arise because people realize that we are in trouble
- Proposition 2: Companies with more volatile results and cash flows will have greater probabilities of bankruptcy for any level of debt and for any level of results.
- Proposition 3: Ceteris Paribus, the higher the overhead costs of bankruptcy, the less debt the company can afford for any level of debt.

Optimal Capital Structure

- ☐ An agency cost comes from hiring someone to do something for us. Emerge therefore our interests may deviate from those of the contracted person.
- ☐ When we lend money to a company, we are allowing stockholders to use that money in the course of managing it. The interests of stockholders are different from ours because
 - ☐ we (like lenders) are interested in recovering the money
 - ☐ Stockholders are interested in maximizing their income
- ☐ In some cases, the set of interests may lead to the stockholders
 - ☐ Investing in more risky projects than desired
 - ☐ To pay themselves high dividends when it would be preferable to keep money in the company.
- ☐ Proposition 4: Ceteris Paribus, the higher the agency problems associated with lending to a company, the less debt the company can afford to afford.

Optimal Capital Structure

- When a company borrows above its capacity, it loses the flexibility of financing future projects with debt.
- So if a company is faced with an unexpected investment opportunity or a business downturn, it will not be able to own debt if it has used too much.
- Proposition 5: Ceteris Paribus, the more uncertain a company is about its future financing needs and projects, the less debt the company will use to finance current projects.

Optimal Capital Structure

- A survey of the Chief Financial Officers of large U.S. companies provided the following ranking (from the most important to the least important) of factors they consider important in their financing decisions
- Factor Ranking (0-5)
 1. Maintain financial flexibility 4.55
 2. Ensuring long-term survival 4.55
 3. Maintain a predictable source of funds 4.05
 4. Maximizing stock prices 3.99
 5. Maintaining financial independence 3.88
 6. Maintain a high debt rating of 3.56
 7. Maintaining the comparison with the pairs 2.47

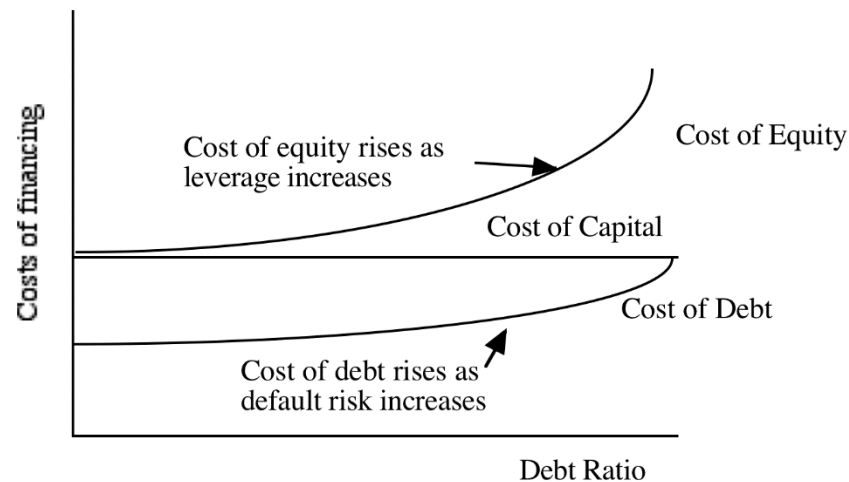
Optimal Capital Structure

<i>Advantages of Debt</i>	<i>Disadvantages of debt</i>
1. Tax Benefit: Interest expenses on debt are tax deductible but cash flows to equity are generally not. <i>Implication: The higher the marginal tax rate, the greater the benefits of debt.</i>	1. Expected Bankruptcy Cost: The expected cost of going bankrupt is a product of the probability of going bankrupt and the cost of going bankrupt. The latter includes both direct and indirect costs. The probability of going bankrupt will be higher in businesses with more volatile earnings and the cost of bankruptcy will also vary across businesses. <i>Implication:</i> 1. Firms with more stable earnings should borrow more, for any given level of earnings. 2. Firms with lower bankruptcy costs should borrow more, for any given level of earnings.
2. Added Discipline: Borrowing money may force managers to think about the consequences of the investment decisions a little more carefully and reduce bad investments. <i>Implication: As the separation between managers and stockholders increases, the benefits to using debt will go up.</i>	2. Agency Costs: Actions that benefit equity investors may hurt lenders. The greater the potential for this conflict of interest, the greater the cost borne by the borrower (as higher interest rates or more covenants). <i>Implication: Firms where lenders can monitor/ control how their money is being used should be able to borrow more than firms where this is difficult to do.</i>
	3. Loss of flexibility: Using up available debt capacity today will mean that you cannot draw on it in the future. This loss of flexibility can be disastrous if funds are needed and access to capital is shut off. <i>Implication:</i> 1. Firms that can forecast future funding needs better should be able to borrow more. 2. Firms with better access to capital markets should be more willing to borrow more today.

Optimal Capital Structure

- In an environment without fees, risk of default or agency costs, the capital structure is irrelevant.
- If Miller Modigliani's theorem is verified:
 - The value of a company will be determined by the quality of its investments and not by its mix of financing.
 - The cost of capital of the company does not change with the debt. As a company raises its debt, the cost of CP increases enough to eliminate any debt gains.

Figure 7.9: Cost of Capital in the MM World



Optimal Capital Structure

- Some authors argue that companies follow a hierarchy of finance, with retained earnings being the most preferred to finance, followed by debt and that new shares issued are the least preferred choice. In particular,
 - Managers value flexibility. Managers value being able to use capital (new investments or assets) without restrictions on the use they give them or having to explain their use to others.
 - Managers value control. They like to be able to keep track of your business.

Rankings	Source	Quotation
1	Retained Results	5.61
2	Simple debt	4.88
3	Convertible debt	3.02
4	External common equity	2.42
5	Simple preferential stocks	2.22
6	Convertible preferential stocks	1.72

Optimal Capital Structure

- How to reach the optimal ratio of capital structure?

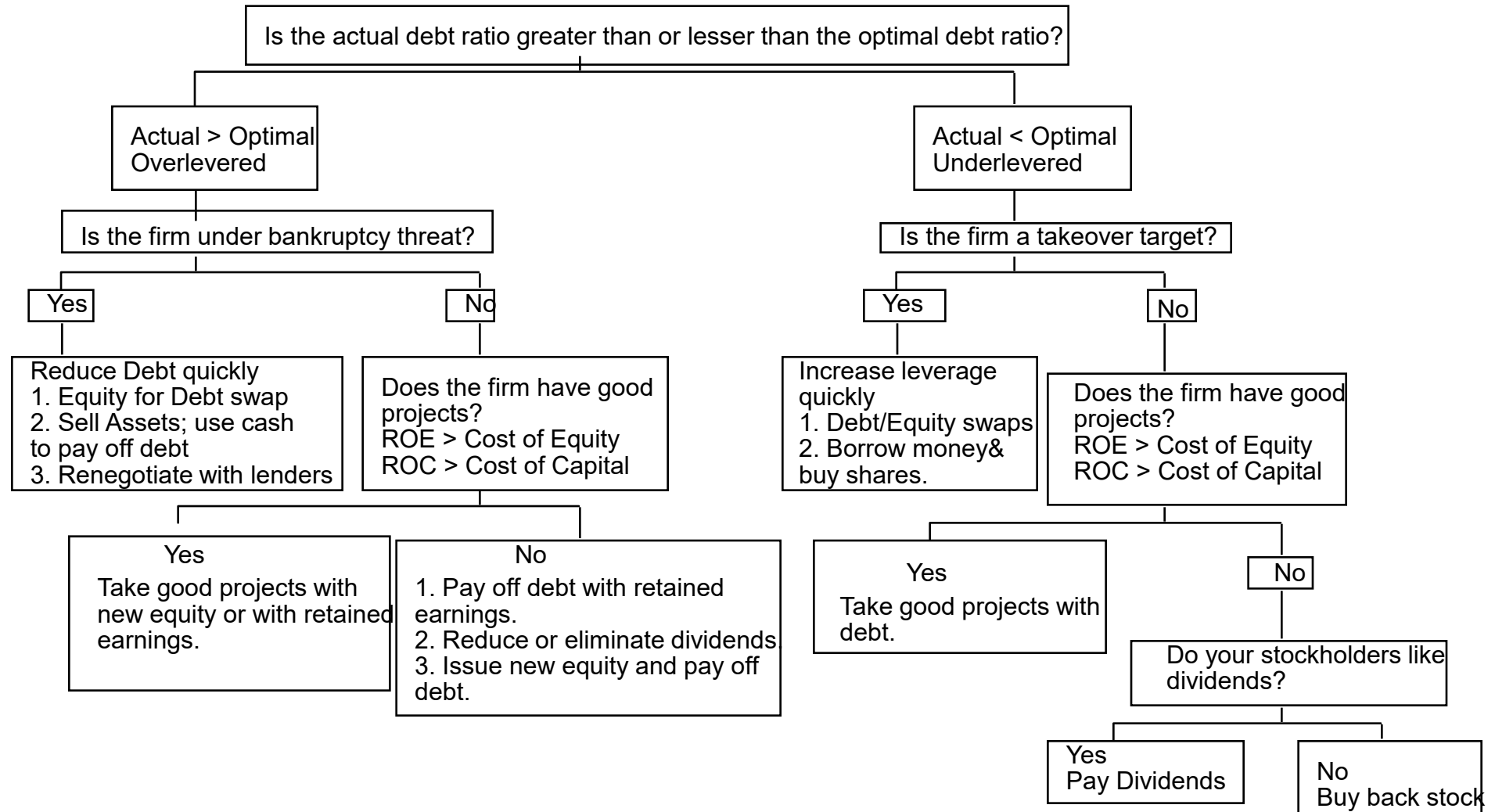
$$\text{Financial Autonomy} = CP / A$$

$$\text{Degree of indebtedness} = P / A$$

$$\text{Financial Structure or debt-to-equity} = P / CP$$

- By the cost of capital: The optimal debt ratio is what minimizes the cost of capital of the company.
- For the cost of reinforced capital: The optimal debt ratio is what generates the best combination of low cost of capital and high operating results.
- Adjusted present value: The optimal debt ratio maximizes the total value of the company.
- By sector: The optimal debt ratio is what brings the company closer to its peers in terms of the financing mix.
- Through the life cycle: The optimal debt ratio is the best fit for the life stage in which the company is in its life cycle.

A Framework for Getting to the Optimal



Empirical determinants of the optimal capital structure (OCS)

- Nelson Vergas, António Cerqueira, Elísio Brandão (2015), The Determinants of the Capital Structure of Listed on Stock Market Nonfinancial Firms: Evidence for Portugal, FEP-UP, School of Economics and Management, University of Porto, Work in Progress => ver artigo Vergas, Cerqueira, Brandão (2015) no elearning
- From the authors abstract:

Empirical determinants of the optimal capital structure

- The study evaluated the effects of debt on the determinants of capital structure, developed by the 4 main lines of thought in this field: trade-off theory, pecking order theory, agency costs theory and market timing theory.
- The sample consisted of non-financial companies listed in the Euronext Lisbon index from 2005 to 2012.
- They used panel data and estimated the fixed effects model.
- The determinants analyzed were: tangibility, profitability, other sources of tax optimization, growth opportunities, size and market valuation.
- The empirical results showed the ability of profitability (-), growth opportunities (+), and other sources of tax optimization (+) in explaining the debt.
- The results confirm the validity of the postulate of pecking order theory.
- Additionally, there are significant changes in the determinants of market valuation, opportunities for growth and tangibility as a result of the 2008 crisis.

Empirical determinants of the optimal capital structure

- The trade-off theory was developed by Kraus and Litzenberger (1973), arguing that firms choose their ideal capital structure by evaluating revenues and costs, debt and equity.
- These authors introduced bankruptcy costs in determining company value. DeAngelo & Masulis (1980) argue that firms seek to achieve the ideal capital structure by considering the benefits and costs of debt in relation to equity - the fiscal savings resulting from the use of debt, combined with the expected bankruptcy costs due to the increase of debt.
- Bradley et al. (1984) conclude that the level of debt is inversely related to the costs of financial risk, including the risk of bankruptcy and agency costs. Based on the trade-off theory, Fama & French (2001) and Beattie et al (2006) determine the optimum level of indebtedness, the interaction between benefits and costs, with the analysis of the effect of an additional unit of debt. Xu Jin (2012), based on the same theory, evaluates the effect of future expectation of return on the capital structure (in debt) of domestic industrial firms in the United States, subject to increased import competition.

Empirical determinants of the optimal capital structure

Jensen & Meckling (1976) developed the study by Modigliani & Miller (1963) and presents the agency cost theory that emphasizes the opportunity cost caused by the impact of debt on company investment decisions, cost monitoring and control with managers and agents and the costs of bankruptcy and reorganization.

Thus, they feel that there are two types of agency costs: agency costs between shareholders and managers and agency costs between shareholders and bondholders. The first ones are related to management control, since the owners of capital seek to ensure that managers act according to their interests, and one of the measures is to increase the responsibility of managers, increasing the level of indebtedness of the company - debt increases decrease the funds available and the possibility for managers to make investments without a positive return or to promote personal compensation (Sayilgan et al, 2006 and Grossman & Hart, 1982).

Empirical determinants of the optimal capital structure

- On the other hand, agency costs between shareholders and bondholders are reflected in the expropriation of shareholder wealth and in their ability to influence the management of the company.
- The existence of agency costs caused by information asymmetry is considered important, and several studies argue that the fact that investors have less information than shareholders, the inflationary effect in the interest rate persists because the investor is more pessimistic (Cortez & Susanto, 2012).

Empirical determinants of the optimal capital structure

- **Myers and Majluf (1984) - the theory of pecking order**
- **This chain does not admit the existence of an optimal capital structure or optimal level of indebtedness, but that companies follow a hierarchical order of reference for debt types - internal resources are first used available and, when insufficient, use of external resources (debt, subordinated debt and last issue of shares).**
- **This order is justified by the absence of additional costs in internally generated funds.**

Empirical determinants of the optimal capital structure

- More recently and with a new approach based on the capital markets, appears the study of **Baker & Wurgler (2002) giving rise to the theory of market timing appears**. These authors consider the capital structure of the companies as a function of the managers' options when looking to make the changes in the share price in the capital market and, thus, optimize the cash inflow - new issues of capital when the stock is overvalued and repurchase when the action is undervalued. **The practice of *market timing* suggests that the choice of optimal moment to issue new shares is the decisive factor in the corporate financing strategy. Funding decisions depend on factors external to the company - such as share appreciation in the market - dependent on the perceptions of the agents: positive or negative expectations of the investors will correspond to times when the company's shares are overvalued or undervalued, respectively. Companies seek to issue new shares when the market value is high relative to accounting value and historical value.**
- The practice of *market timing* by the companies was evident in the work of Frankel & Lee (1998) and La Porta (1996) when they studied the growth opportunities and their relationship (inverse) to the profitability of actions and relationship with the expectation of investors. Finally, **Aydogan Altı (2006) evaluates the effects of the market timing theory for the non-financial companies in the USA and confirms the negative correlation between debt and market overvaluation.**

Empirical determinants of the optimal capital structure

- The dependent variable in the study is the company's level of indebtedness.
- Xu Jin (2012) and Rogão & Serrasqueiro (2008) identified the debt by the Total Liabilities / Total Assets ratio.
- Rajan & Zingales (1995) and Harris & Raviv (1991) used Total Liabilities / (Assets - Cash - Other debtors).
- Padron et al. (2005) used the Total Liabilities / Capital Market Value ratio
- (2005) based on Gaud et al. (2005), used the Total Liabilities / Over Total Equity ratio
- In this study they used: LEV_A where $LEV_A_{i,t} = \text{Total Liabilities}_{i,t} / \text{Total Assets}_{i,t}$ of the company i in year t

Empirical determinants of the optimal capital structure

- The literature presents several determinants of the capital structure.
- This paper analyzes the effects of six determinants of capital structure on firm debt: tangibility, profitability, other sources of tax optimization, size, growth opportunities and the market valuation of the company.

Empirical determinants of the optimal capital structure

Tangibility

The tangible assets of a company are considered one of the major guarantees for creditors and its importance in the capital structure of the company has an increasing importance in debt (Padron et al, 2005). Sayilgan et al. (2006) and Gaud et al. (2005) added inventories to fixed assets because in many situations stocks have a significant value at a time of settlement

Here they use the variable (TANG_I):

$$\text{TANG_I}_{i,t} = (\text{Fixed Assets} + \text{Inventory})_{i,t} / \text{Total Assets}_{i,t}$$
 of the company i in year t

Optimal Capital Structure

Profitability

Titman & Wessels (1988) defined the variable as the operating profit on sales or operating results of the asset. Sayilgan et al. (2006), Rajan & Zingales (1995) and Myers (1984) defined the variable Return on Assets (ROA, calculated by the EBITDA ratio over total assets). In the article use variable profitability (PROF):

$$\text{PROFI, } t = \text{EBITDA, } t / \text{Total Assets, } t \text{ of the company } i \text{ in year } t$$

Optimal Capital Structure

Other sources of fiscal optimization

Musulis & DeAngelo (1980) characterize the rate optimization by depreciation and amortization when they do not consider financial charges.

In this article, we use the interpretation of Cortez & Susanto (2012), Sayilgan et al (2006) and Titman & Wessels (1988) as measured by total depreciation and amortization over total assets (NTDS):

$NDTS_{i,t} = \text{Depreciation and Amortization, } t / \text{Total Assets, } t \text{ of the company } i \text{ in year } t$

Optimal Capital Structure

Dimension

The size of the company is an indicator commonly used to explain the levels of indebtedness and the ability of companies to obtain new financing in the market. Larger companies are more stable, less volatile and can exploit economies of scale (Gaud et al, 2005 and Graham et al, 1998).

Larger companies can obtain lower financial costs because they have lower default risks and size is a good proxy for the probability of default (Rajan & Zingales, 1995).

In the literature, we use different indicators of size such as the logarithm of net sales (Cortez & Susanto, 2012; Sayilgan et al., 2006, Gaud et al., 2005, Titman & Wessels, 1988 and Rajan & Zingales, 1995). total assets (Padron et al, 2005).

In this work the authors use (SIZE):

$SIZE_{i,t} = \ln \text{Net Sales}_{i,t}$ of the company i in year t

Optimal Capital Structure

Opportunities for growth

In the literature the growth opportunities are related to the new investments and the market evaluation of the companies. Some indicators such as the company's market value ratio over total liabilities (Padron et al, 2005), the annual growth of total assets or the total fixed assets of companies (Cortez & Susanto, 2012, Sayilgan et al, 2006 and Titan and Wessels, 1988), the ratio of investment expenditure to total assets (Titman & Wessels, 1988), the ratio of market value to book value of assets (Gaud et al., 2005, Rajan & Zingales, 1995 and Myers, 1977) are considered viable to measure the effects of growth opportunities.

Here they use the variable (GRA): $GRA_{i,t}$ = Annual Growth of Total Assets, t of the company i in year t

Optimal Capital Structure

Market valuation

The work of Baker & Wurgler (2002) proposes the market-to-book ratio as a proxy in the analysis of the company's relationship with the practice of debt market timing in the stock market and assess the impact of short- and long-term effects on the structure of capital

They show the sustainability of this ratio in the analysis of its effects on company debt and to test the theory of market timing, the authors in the present study use the MTB: $MTB_{i,t} = \text{Market Value of Assets}_{i,t} / \text{Book Value of Assets}_{i,t}$, of the company i in year

Where the market value of assets = Total assets - Value Common shares + Market capitalization and the book value of assets = total assets.

Tested Hypothesis

- **H1 - Tangibility has a positive impact on indebtedness**
- (2004), and Xan Jin (2012) and Rajan & Zingales (1995), based on the trade-off theory, find a positive relationship between the two, which is due to the existence of superior collateral over assets. Companies with high fixed asset needs have higher borrowing requirements and therefore the debt level tends to be higher (Rajan & Zingales, 1995; Harris & Raviv, 1991).
- Based on the pecking order theory, Gaud et al (2005) confirm that tangible assets have a positive impact on management decisions on financing, since they are less subject to information asymmetry problems and reduce credit risk (they have a higher value in the case of bankruptcy) - the higher the value of tangible assets, the higher the level of indebtedness as there are more guarantees to the loan.
- According to the agency costs theory, debt plays a disciplining role in managers because it reduces available cash flows (Harris & Raviv, 1991 and Grossman & Hart, 1982) and tangible assets reduce agency costs by allowing debt to be increased by using assets as a guarantee (Cortez & Susanto, 2012).

Tested Hypothesis

- **H2 - Profitability has a negative impact on debt**
- Fama & French (2002) compare the trade-off and pecking order theories concluding that companies with higher rates, more profitable and with reduced profit volatility have a higher level of indebtedness.
- The trade-off theory considers that decisions about indebtedness are influenced by fiscal benefits and that high levels of results influence the ability to obtain financing in the market, with a positive relation between profitability and debt (Rogão & Serrasqueiro, 2008).
- Gaud et al (2005) argue that good results in the past are a good proxy for future earnings expectations, concluding that more profitable firms can increase affordability (Sayilgan et al, 2006).
- However, Jin Xu (2012) tested the postulate of the trade-off theory where they find evidence of a negative relationship between profitability and indebtedness. Based on the pecking order theory, Nakamura et al. (2007) and Rajan & Zingales (1995) find evidence of a low level of indebtedness in the most profitable companies justified by the fact that high performance companies prefer the use of internal resources to finance their projects.

Tested Hypothesis

- Finally, in the agency costs theory, the level of increasing indebtedness provides the shareholders with mechanisms to monitor and control the available cash flow problem, where financing is a way to reduce the available financial resources and to reinforce the accounting of the managers in the development of new projects and not to pursue individual goals (Jensen, 1986).
- Also, Grossman & Hart (1982) argue that agency costs can be reduced by reducing liquidity and dividends only by increasing indebtedness in the firm's capital structure and by considering information asymmetry, concluding that there is a relationship between profitability and debt.

Tested Hypothesis

- **H3 - The tax benefit of non-debt has a negative impact on the indebtedness**
- The trade-off theory considers depreciation and amortization in fiscal management as a direct substitute for the burden associated with indebtedness (Cortez & Susanto, 2012). Miguel & Pindado (2001) conclude that companies with high levels of depreciation and amortization have a lower level of indebtedness in their capital structure.
- On the other hand, Titman & Wessels (1988) obtained different conclusions because they do not find statistical significance.
- Bradley et al. (1984) confirm the positive effect in companies with higher levels of asset investment. Graham (2005) explains the positive relationship shown by the investment-to-return ratio because the most profitable companies make more investments through external financing, with a positive relationship between depreciation and amortization and indebtedness.
- Cortez & Susanto (2012), Sayilgan et al (2006) and Miguel & Pindado (2001) confirm the existence of a negative effect on the non-financial sector

Tested Hypothesis

- **H4 - The size of the company has a positive impact on the indebtedness**
- By trade-off theory there is a positive relationship between size and debt. Graham et al. (1998) argue that large firms tend to go lower in bankruptcy, and thus obtain financing faster and Sayilgan et al. (2006) argue that large firms are government-protected which allows them to take a higher risk by obtaining more indebtedness.
- Based on the pecking order theory, Harris & Raviv (1991) consider the existence of a positive relation because the big companies provide more information to the market thus obtaining greater confidence and greater openness to new financing by the creditors.
- (2002) and Rajan & Zingales (1995) obtained inconclusive (mixed) results, but the results showed that there was a negative relationship between debt level and size, which attribute to the existence of ambiguity of the scale effect (large companies use a greater diversity of financing - external and internal).

Tested Hypothesis

- **H5 - Growth opportunities have a positive effect on indebtedness**
- By trade-off theory, high-growth companies have lower indebtedness because companies and lenders have less propensity for new loans - vulnerability, cost, and risk is higher in these projects because of greater uncertainty (Cortez & Susanto, 2012). Gaud et al. (2005), Fama & French (2002), Rajan & Zingales (1995) and Titman & Wessels (1988) conclude that there is a negative relationship between growth opportunities and debt levels.
- On the other hand, Jensen (1986) supports the theory of agency costs to conclude that greater growth opportunities lead to more indebtedness to minimize agency costs between managers and shareholders because they use debt to discipline managers.
- Based on the pecking order theory, firms with higher growth opportunities need higher funding amounts, encouraging managers to obtain external funding sources and generating a higher return to creditors (Song, HS, 2005).

Tested Hypothesis

- **H6 - Stock market valuation has a negative effect on indebtedness**
- Baker & Wurgler (2002) conclude that firms tend to increase funds available through new equity issues when market value is high and firms increase borrowing when their market valuation is low. Companies with a high market to book ratio reduce their level of indebtedness, and companies with low market to book ratio tend to increase their debt rather than using the capital market.
- Rajan and Zingales (1995) conclude that there is a negative relationship between growth opportunities (the proxy used was the MTB) and the indebtedness arguing that companies choose to issue new shares to finance themselves, especially when the market value of the company is high (overvalued) and thus reduce the level of indebtedness.

Data

Table 1 - Definition of independent variables and expected signs

Hypotesys	Variable	Definition	Trade-Off Theory	Pecking Order Theory	Agency Costs Theory	Market Timing Theory	Expected Sign
H1	TANG_I	Tangible Assets plus Inventories / Total Assets	+	+	+		+
H2	PROF	Earnings before interest, tax and depreciation / Total Assets	+/-	-	+		-
H3	NDTS	Annual Depreciation Expense / Total Assets	-				-
H4	SIZE	Log (Net Sales)	+	+/-			+
H5	GRA	Percentage change in Total Assets	-	+	+		+
H6	MTB	Market-to-book (market value / book value)				-	-

Data / Model

$$\text{LEV_A}_{i,t} = B0 + B1*\text{TANG_I}_{i,t} + B2*\text{PROF}_{i,t} + B3*\text{NDTS}_{i,t} + B4*\text{SIZE}_{i,t} + B5*\text{GRA}_{i,t} + B6*\text{MTB}_{i,t} + U_i + V_t + E_{i,t} \quad (3.1)$$

Where "i" represents the individual companies and "t" the year, we added the *dummy* variables, "U" and "V", incorporating the fixed effects of the companies (cross-section) and of the years (time series), respectively; "E" represents the error of the model designated as disturbance term. We estimate by the method of least squares (OLS) with fixed effects, applied to the panel data and admit the existence of fixed, unobservable effects for the companies, individually, and for years (details attached).

Table 2 – Business industry

Type of companies	Number of companies (Start)	Number of companies (End)
Wholesale trade	8	7
Retail trade	2	2
Construction and real estate	5	5
Electric power and gas	3	3
Hotel and restaurants	2	2
Manufacturer and machinery	10	10
IT and services	2	2
Sports institutions	3	3
Banks, securities and financing	3	0
Information and communication	6	6
Transportation	1	1
TOTAL	45	41

Results

Table 6 – Equation (3.1) estimation results I

Variable	Expected Sign	2005 - 2012
TANG_I	+	0,103225 (1.458128)
PROF	-	-0,42983*** (-3.431451)
NDTS	-	3,088476*** (4.749085)
SIZE	+	-0,014262 (-0.575214)
GRA	+	0,054899** (2.203533)
MTB	-	0,02049 (0.841173)
Number of observations		277
F statistic		18.18936***
R ²		0,808522

Table 8 – Equation (3.1) estimation results II

Variable	Expected Sign	2005 - 2008	2009 - 2012
TANG_I	+	0,148248 (0.870863)	0,170931** (2.262815)
PROF	-	-0,313909** (-2.377269)	-0,417606** (-2.323114)
NDTS	-	0,517456 (0.472131)	1,456597 (1.534897)
SIZE	+	-0,022064 (-0.429534)	0,005353 (0.188771)
GRA	+	0,063642** (2.398915)	0,069014 (1.433504)
MTB	-	-0,10934*** (-3.604427)	0,206216*** (4.986021)
Number of observations		116	161
F statistic		16.66825***	26.34627***
R ²		0,922729	0,920826

This table reports OLS estimation results on equation (3.1), as well redundant fixed effects tests. *, ** and *** denote coefficients significance at 10%, 5% and 1% levels, respectively. Values below coefficients are corresponding t-statistics. The panel sample covers the period from 2005 to 2012. Consult the table 1 for variable definitions.

Results

Table 7 – Comparison of the test results with the expectations theories I

Variable	Trade-Off Theory	Pecking Order Theory	Agency Costs Theory	Market Timing Theory	Expected Sign	Results 2005-2012
TANG_I	+	+	+		+	+
PROF	+	-	+		-	****
NDTS	-				-	****
SIZE	+	?			+	-
GRA	-	+	+		+	***
MTB				-	-	+

*, ** and *** denote coefficients significance at 10%, 5% and 1% levels, respectively

Table 9 – Comparison of the test results with the expectations theories II

Variable	Trade-Off Theory	Pecking Order Theory	Agency Costs Theory	Market Timing Theory	Expected Sign	Results 2005 - 2008	Results 2009 - 2012
TANG_I	+	+	+		+	+	***
PROF	+	-	+		-	**	**
NDTS	-				-	+	+
SIZE	+	?			+	-	+
GRA	-	+	+		+	***	+
MTB				-	-	****	****

*, ** and *** denote coefficients significance at 10%, 5% and 1% levels, respectively

Conclusions

The estimation results show the supremacy of the *pecking order* theory, in the choices of managers, with relevance and significance in the determinants of profitability (-) and growth opportunities (+) as main factors of corporate debt. Additionally, the changes in the determinants of market valuation and tangibility are highlighted, as the main effects of the financial crisis in the period under analysis; statistical tests can validate hypotheses H2 and H5. However, with the division of the sample, it was possible to validate H6 in the period before the crisis and H1 in the post crisis period.

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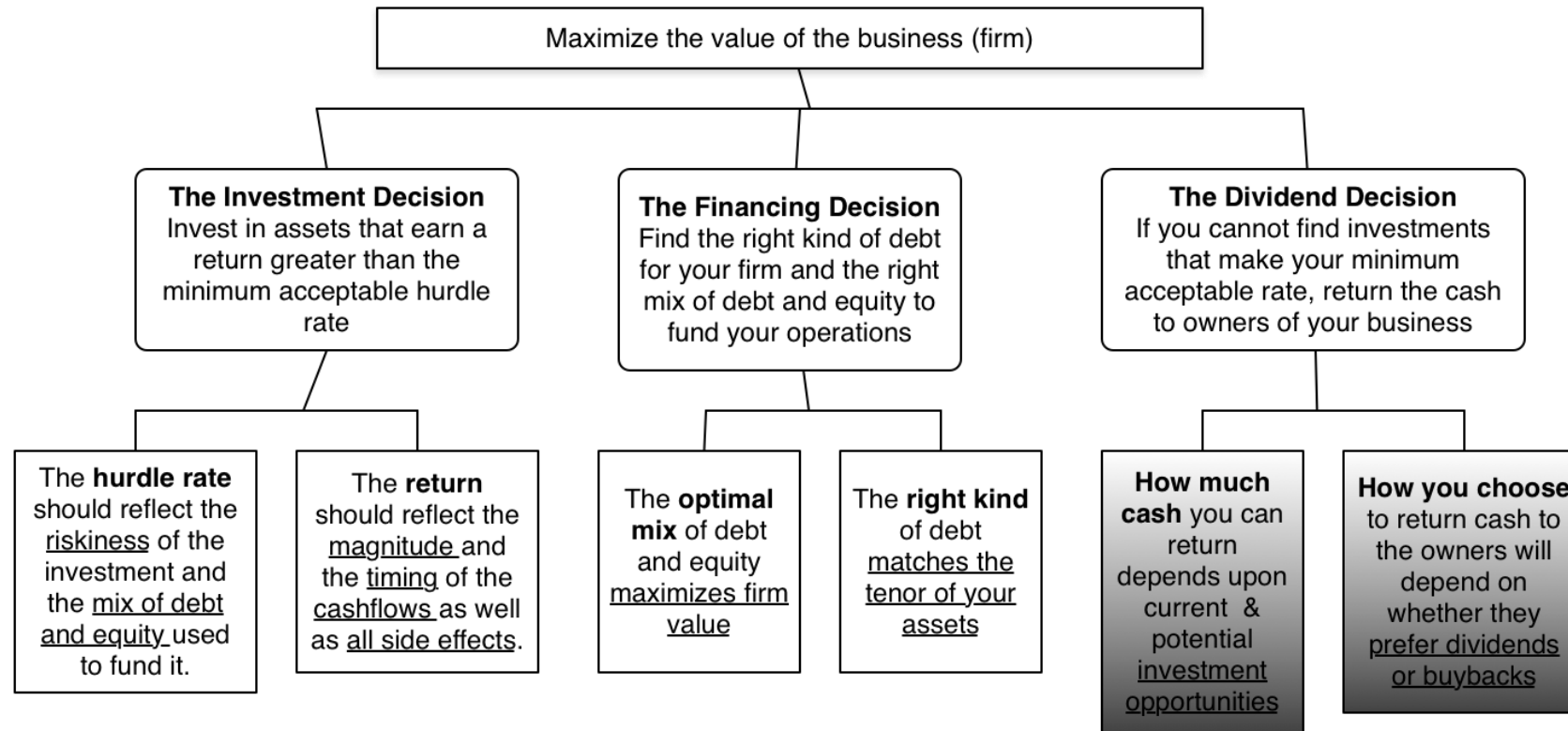
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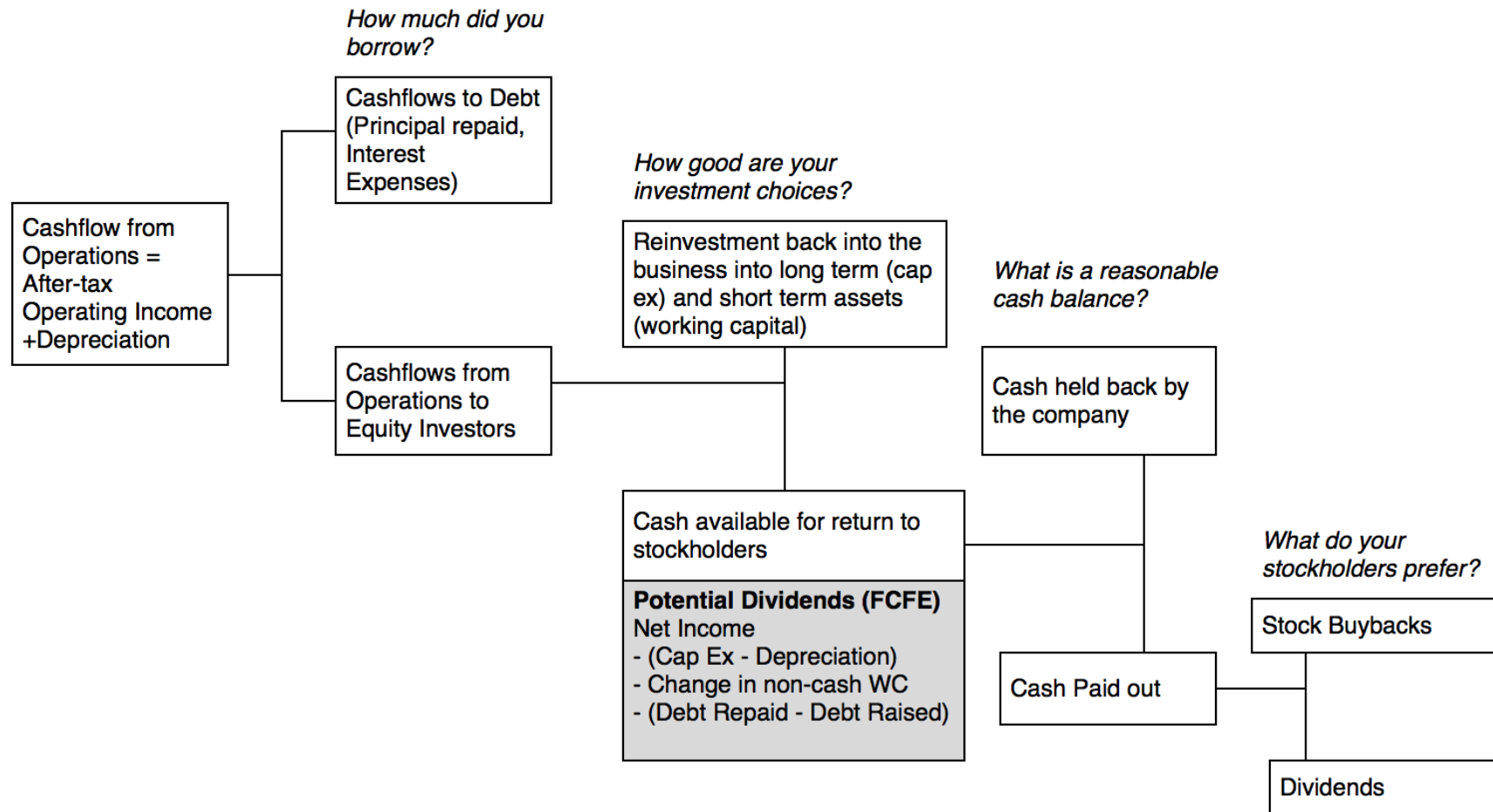
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Dividend Policy



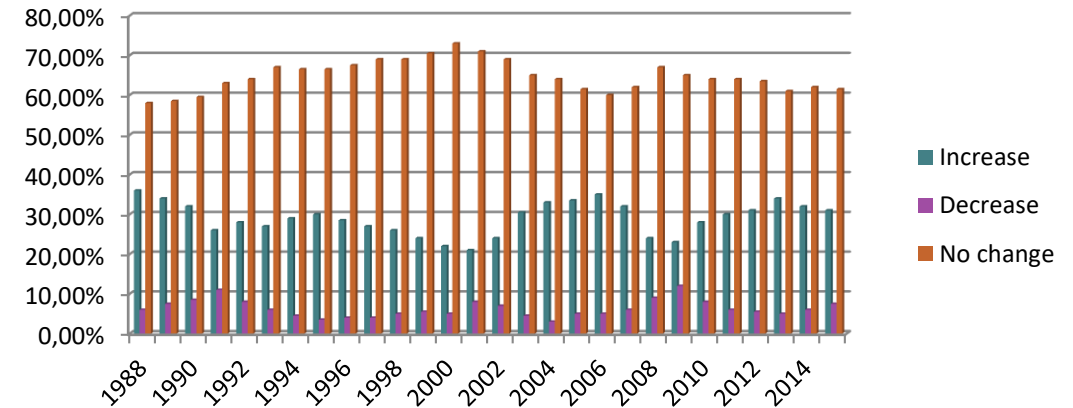
Dividend Policy



Dividend Policy

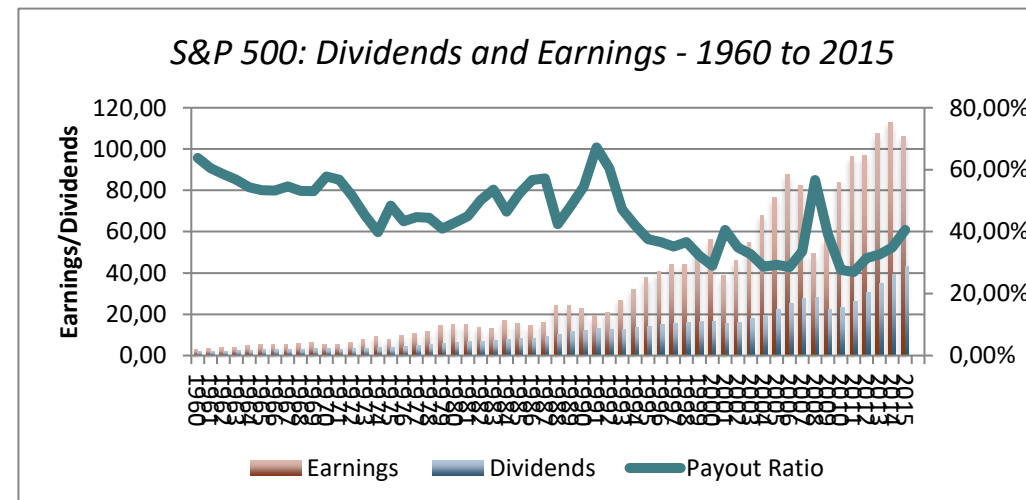
- I. Dividends are sticky

Dividend Changes at US companies



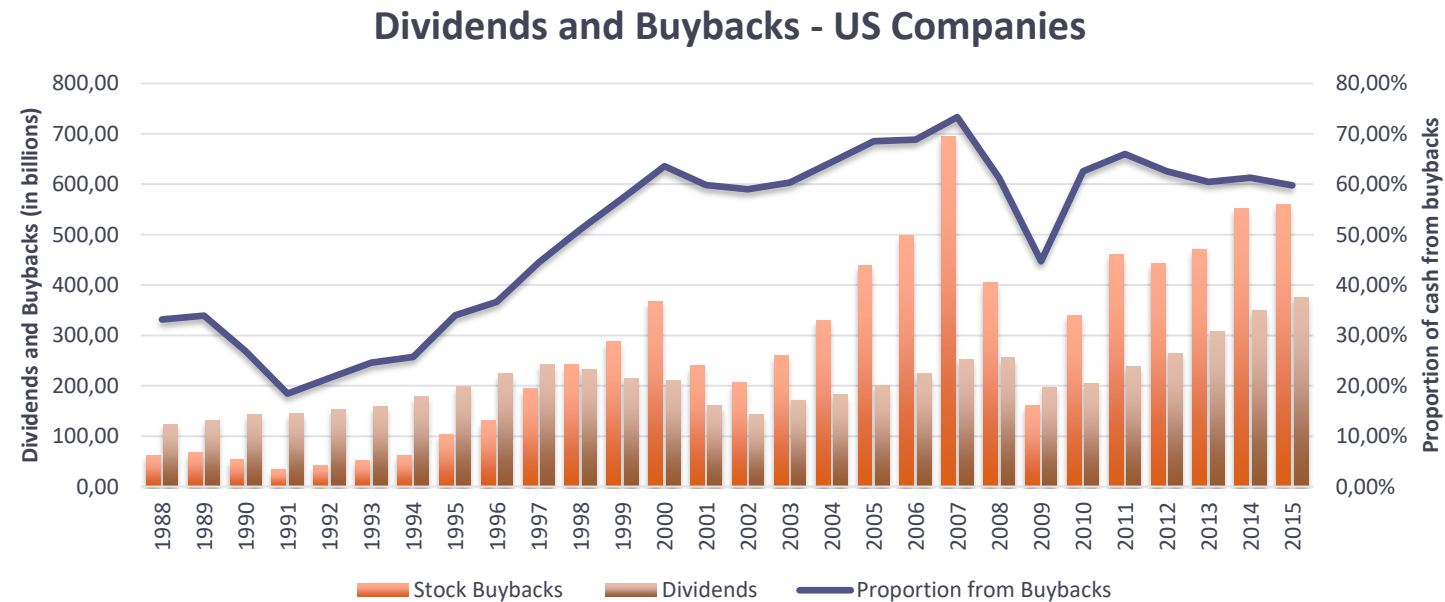
- II. Dividends tend to follow earnings

S&P 500: Dividends and Earnings - 1960 to 2015



Dividend Policy

- III. Are affected by tax laws...
- IV. More and more US firms are buying back stock, rather than pay dividends...



Measures of Dividend Policy

- Dividend Payout = Dividends / RL = (1-b)
 - It measures the % of results that the company pays on the form of dividends
 - If the RL is negative, the payout ratio can not be calculated.
- Dividend Yield = Dividends per share / Share Price or Quotation
 - Measures the return that the investor can obtain from dividends
 - It becomes part of the expected return on investment.

3 schools of thought about Dividend Policy

1. If there are no disadvantages of fees associated with dividends and companies can issue shares without issuing costs to increase the CP whenever necessary
2. Dividends do not matter and dividend policy does not affect the value
3. If dividends create a disadvantage of rates for investors (relative to capital gains)

Dividends are bad and increasing dividends reduces the value

If dividends create a tax advantage for investors (relative to capital gains) and / or shareholders like dividends

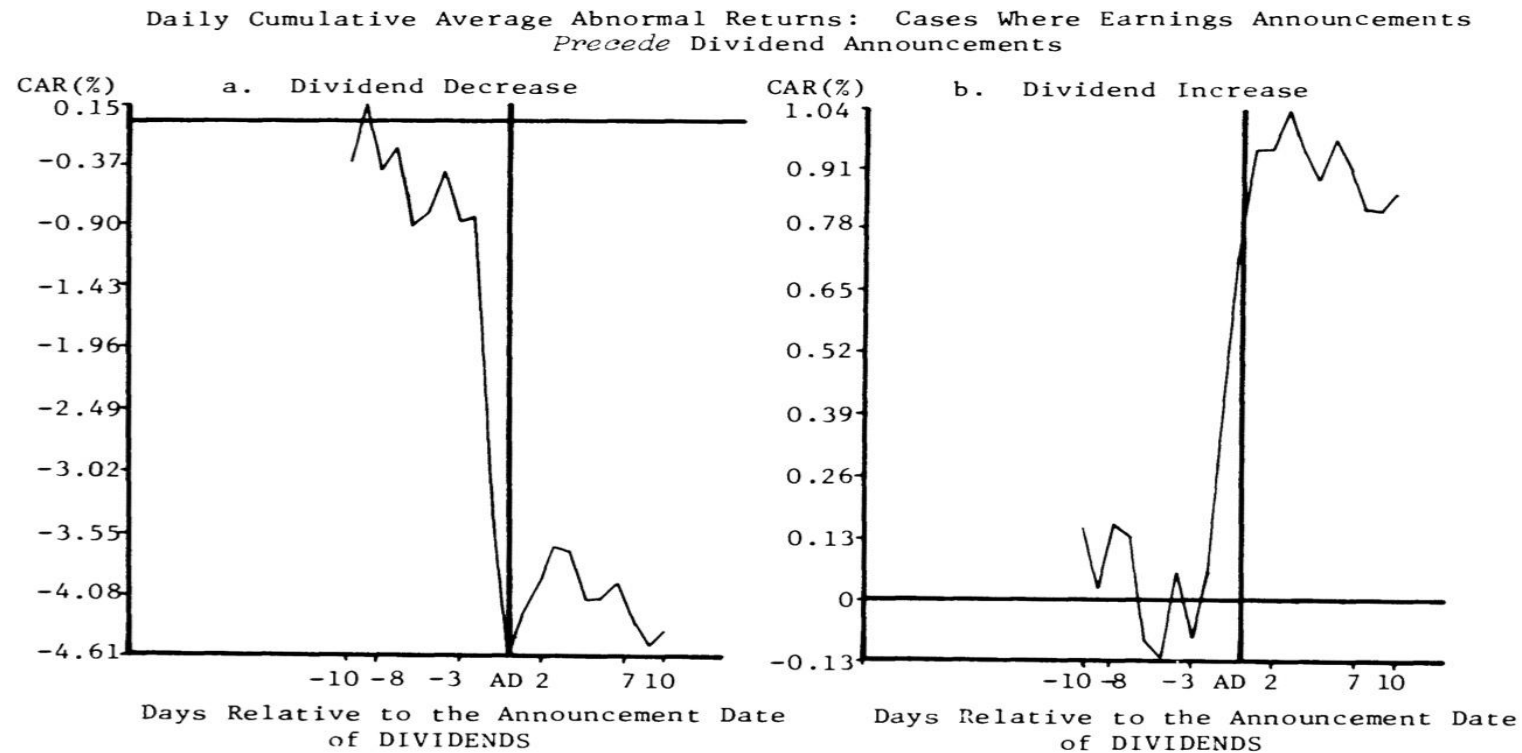
Dividends are good and increasing dividends will increase the value

Point of view of the "balance" required:

1. If a company has too much money and few good investment opportunities ($NPV > 0$), returning money to shareholders (dividends or share repurchase) is good.
2. If a company does not have excess cash and / or has many good investment opportunities ($NPV > 0$), returning money to shareholders is bad.

Dividend Policy

- Dividends send a signal: Dividend increases are good news ...



Dividend Policy

- What managers believe about dividends (survey):

<i>Statement of Management Beliefs</i>	<i>Agree</i>	<i>No Opinion</i>	<i>Disagree</i>
1. A firm's dividend payout ratio affects the price of the stock.	61%	33%	6%
2. Dividend payments provide a signaling device of future prospects.	52%	41%	7%
3. The market uses dividend announcements as information for assessing firm value.	43%	51%	6%
4. Investors have different perceptions of the relative riskiness of dividends and retained earnings.	56%	42%	2%
5. Investors are basically indifferent with regard to returns from dividends and capital gains.	6%	30%	64%
6. A stockholder is attracted to firms that have dividend policies appropriate to the stockholder's tax environment.	44%	49%	7%
7. Management should be responsive to shareholders' preferences regarding dividends.	41%	49%	10%

Determinants of Dividend Policy

- Yusniliyana Yusof, Suhaiza Ismail , (2016),"Determinants of dividend policy of public listed companies in Malaysia", Review of International Business and Strategy, Vol. 26 Iss 1 pp. 88 – 99 Permanent link to this document: <http://dx.doi.org/10.1108/RIBS-02-2014-0030> - see article YusofIsmail2016
- P.G.Thirumagal *et al.*, *LITERATURE REVIEW ON FACTORS INFLUENCING DIVIDEND DECISIONS*, International Journal of Engineering, Business and Enterprise Applications, 11(1), December 2014-February 2015, pp. 48-52 – see article Thirumagal 2015

Dividend Policy

- **From the first article:**
- **Purpose** – This study aims to investigate the determinants of the dividend policy of public listed companies in Malaysia. The factors examined in this study include earnings, cash flows, free cash flows, debt level, growth, investment, size, largest shareholders, risk and lagged dividend.
- **Design/methodology/approach** – Data were obtained from the relevant databases and annual reports of the sampled companies. The study examines a total of 147 listed companies. In analyzing the data, the study used fixed and random effects, pooled least squares model, robust standard errors on fixed effects and random-effects models.
- **Findings** – The results revealed the five factors that are earnings, debt, size, investment and largest shareholder have a significant influence on dividend policy, with earnings, firm size and investment revealed to have a positive significant effect, while debt and large shareholders have a negative significant effect.

Factors which influence DP

- **Beta:** *Alli, Khan and Ramirez (1993) found that dividends do not provide information on a company's future cash flows. They conclude that to the beta, the company's capital expenditures and financial backlash are inversely related to the payout dividend.*
- **Liquidity:** *Kanwal and Kapoor (2008) examined PD and explore factors such as profitability, cash flows, tax rate, sales growth and growth opportunities, all of which have an influence on the company's PD. They conclude that cash flows that indicate liquidity and the beta that indicates risk are the strongest determinants.*

Factors which influence DP

- **Profitability:** Factors such as profitability, risk, cash flows, agency costs, and growth affect the payout ratio (see Higgins 1981, Rozeff 1982, Luit et al., 1985, Pruitt and Gitman 1991, Jensen et al., 1992 Ali et al., 1993, Collins et al., 1996, D'Souza, 1999).
- The literature states that profitability is a significant variable in dividend policy (Jensen et al., 1992; Han et al., 1999; Fama and French, 2000). However, there is a significant difference between the developed and developing countries' dividend policy that was identified by Glen et al. (1995), which showed that dividend payment rates in developing countries are 2/3 of developed countries. Companies in emerging countries do not follow a stable distribution policy, and the distribution rate depends on the company's profitability in that year.
- Profitability (PROF) is the RL / Amount ratio that shareholders have placed in the company. ROE has been used in many studies as a proxy for profitability (Aivazian et al., 2003, ap Gwilym et al., 2004) and is calculated as: $PROF = (\text{Net profit} / \text{shareholder's equity}) * 100$.
- Pruitt and Gitman (1991), in their study verify that the present and past results are important factors in the decision to pay dividends as well as Al Kuwari (2009) also found a positive relationship between the two.

Factors which influence DP

- **Earnings:** The empirical analysis of Adaoglu (2000) shows that companies follow unstable dividend policies and that the most determinant factor is their earnings (one of the determinants of PD is the anticipated future earnings).
- Eriotis (2005) states that the dividend distribution rate is determined by distributed earnings and size. The results of Raj kumar & Pawan kumar jha (2012) show that post-tax RL, cash flows and amortization have an impact on dividends.
- Myers (2004) finds strong evidence for earnings, profit margin, property, and the debt-equity ratio on the dividend decision.
- Kevin (1992) discovers that profitability and earnings are the two most important actors in determining dividends.

Factors which influence DP

- **Firm size:** There is a significant and strong association between the decision to pay dividends and the size of the company, profitability, growth, indebtedness, balance and historical dividends (DeAngelo et al., 2004).
- The research by Lloyd, Jahera, and Page (1985), and Vogt (1994) indicates that the size of the firm plays a role in explaining the dividend payment ratios (1-b). They find that larger firms tend to be more mature and soon to have easier access to capital markets, reducing their reliance on domestic funds, and allowing for higher dividend payment ratios. The relationship between the dividend payout ratio and the size is expected to be positive.
- The variable (SIZE) is measured as the natural logarithm of total assets due to the fact that larger firms will pay higher dividends to reduce agency costs (Ghosh and Woolridge 1988, Eddy and Seifert 1988, Redding 1997) .
- Eddy and Seifert (1988), Jensen et al. (1992), Redding (1997), and Fama and French (2000) also indicate that larger firms distribute more dividends on the monetary form.

Factors which influence DP

- **Growth:** Naceur et al. (2006) find that the most profitable companies with more stable results tend to pay higher dividends. Also, fast-growing companies also distribute more dividends to attract investors. D'Souza (1999) shows a positive but not significant result in the case of growth.
- Higgins (1972) shows that the payout ratio is negatively related to the need for a company's funds to finance growth opportunities.
- Rozeff (1982), Lloyd et al. (1985), and Collins et al. (1996) all show a negative and significant relationship between sales growth and dividend payout.
- The growth rate is measured as the rate of growth of sales (Rozeff, 1982; Lloyd et al., 1985, Jensen et al., 1992, Alli et al., 1993; Mohd et al., 1995; Holder et al. (1998), Chen et al. (1999) and Sexsena (1999).
- Therefore, the literature identifies positive and negative relationships between the distribution of dividends and the growth of sales, that is, contradictory results

Factors which influence DP

- **Corporate tax:** *Omet (2004) concludes that the dividend tax rate has no significant impact on the dividend behavior of listed companies.*
- *The corporate tax rate was negatively associated with the payout dividend by Anil and Kapoor (2008) that they found to be insignificant. Mahira Rafique (2012) conclude that the tax rate and size have a significant relationship with the dividend payout.*
- **Cash flow:** Alli et al. (1993) show that dividend payments depend more on cash flows that reflect the company's ability to pay dividends, rather than on current earnings, since they are less influenced by accounting practices. They argue that current results do not reflect the ability to pay dividends. The proxy they use for earnings is the ratio $(EBIT = \text{earnings before interest and taxes}) / \text{total assets}$.
- Brittain (1966) studied the rate structure and PD concluding that the ability of companies to pay dividends is best explained when considering cash flows.

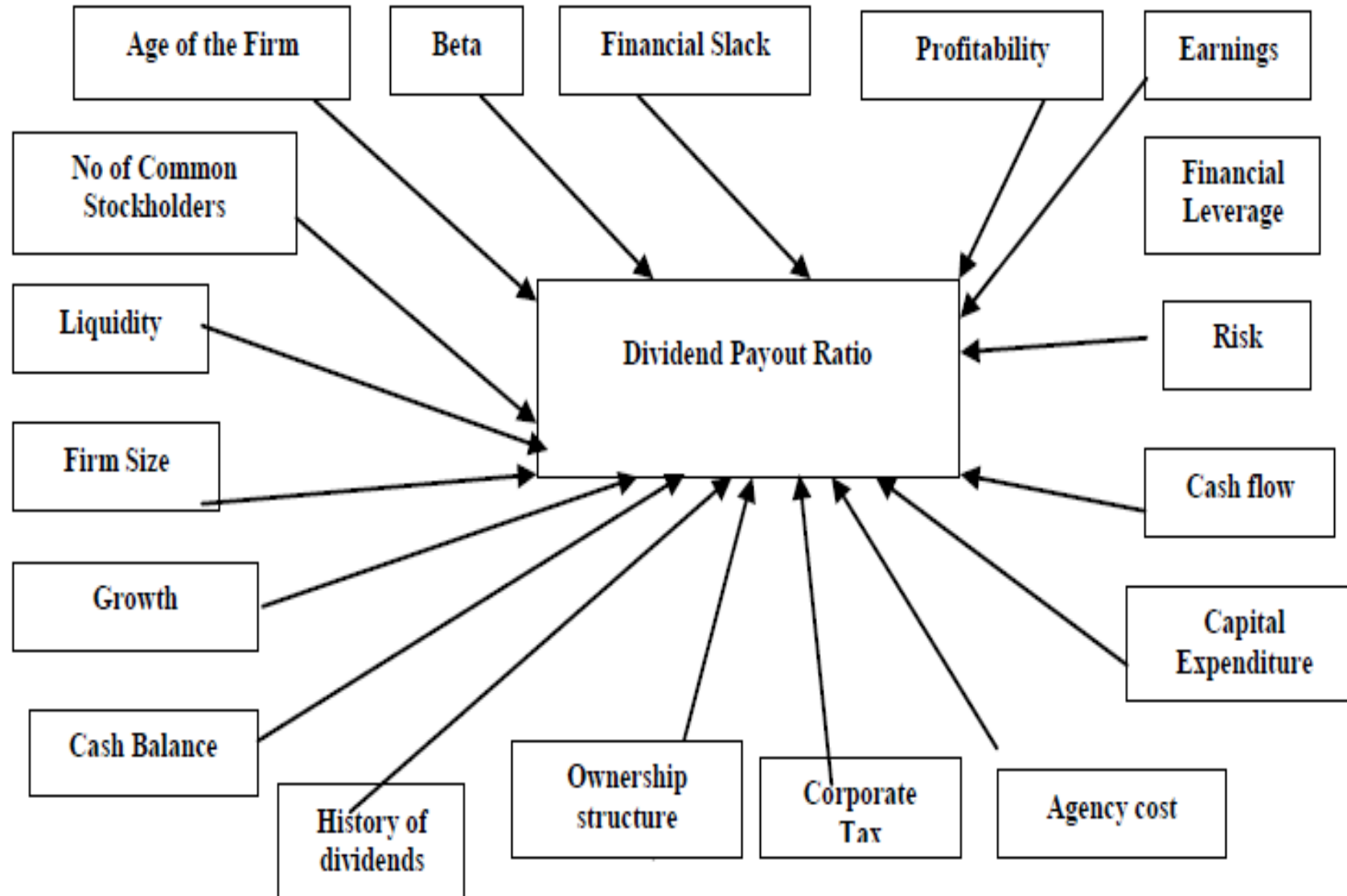
Factors which influence DP

- **Financial leverage:** Some studies have concluded that the level of indebtedness negatively influences the PD (Jensen et al., 1992; Agrawal and Jayaraman, 1994; Crutchley and Hansen, 1989; Faccio et al., 2001; Gugler and Yurtoglu, 2003; Al-Malkawi, 2005).).
- They conclude that more indebted companies tend to retain results in order to be able to have domestic financing rather than distribute it to shareholders and thus also protect creditors. Mollah et al. (2001) conclude that firms with high debt ratios have high transaction costs and are in a weak position to pay higher dividends to avoid the cost of external financing. Usually the financial leverage ratio is used, or debt-to-equity ratio (short-debt + long-term debt) / CP ratio. Al Kuwari (2009) also found a negative relationship between the two: PD and EF.
- Amithabh Gupta & Charu Banga (2010) find that indebtedness and liquidity are the two factors that most affect PD or dividend decisions.

Factors which influence DP

- **Capital Expenditure:** There is a negative relationship between agency costs and market risk with the dividend payout ratio but not between dividend payout and investment opportunities (D'Souza, 1999)
- **History of dividends:** Naeem and Nasr (2007) observed determinants and trends of PD. They show that companies are either reluctant to pay dividends or pay them very low since dividends and current dividend decisions depend on the previous year and the profitability ratio.
- **Ownership structure:** Husam et al (2007) examined the determinants of corporate PD and show that the proportion of shares held by insiders and their holdings by the state significantly affect the amount of dividends paid. Kania and Bacon (2005) find that variables such as sales growth, expansion and ownership structure have a negative impact on dividend decisions.

Factors which influence DP



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- <https://www.mdpi.com/1911-8074/18/2/70>
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- <https://ideas.repec.org/a/bjx/jomwor/v2024y2024i4p549-561id460.html>

Additional Notes OCS - Optimal Capital Structure

Appendix

Additional notes: Optimal capital structure

- How to reach the optimal ratio of capital structure?

$$\text{Financial Autonomy} = CP / A$$

$$\text{Degree of indebtedness} = P / A$$

$$\text{Financial Structure or debt-to-equity} = P / CP$$

- By the cost of capital: The optimal debt ratio is what minimizes the cost of capital of the company.
- For the cost of reinforced capital: The optimal debt ratio is what generates the best combination of low cost of capital and high operating results.
- Adjusted present value: The optimal debt ratio maximizes the total value of the company.
- By sector: The optimal debt ratio is what brings the company closer to its peers in terms of the financing mix.
- Through the life cycle: The optimal debt ratio is the best fit for the life stage in which the company is in its life cycle.

Optimal Capital Structure

- Company value = Present value of the company's cash flows, discounted to cost of capital.
- If the company's cash flows remain constant and the cost of capital is minimized, the company's value will be maximized.
- 1) By the cost of capital:
- The cost of debt is the market interest rate that the company has to pay on the long-term loan it took today, net of tax benefits. It will be a function:
- (a) the risk-free long-term interest rate
- (b) The company default spread, which reflects your credit risk
- (c) The company's marginal tax rate
- The cost of equity reflects the expected return required by equity investors. If they are diversified, only the portion of CP risk that can not be diversified (beta) will be "weighted" in the cost of that capital.
- The average cost of capital (WACC) will be the cost of each capital component weighted by its relative market value.

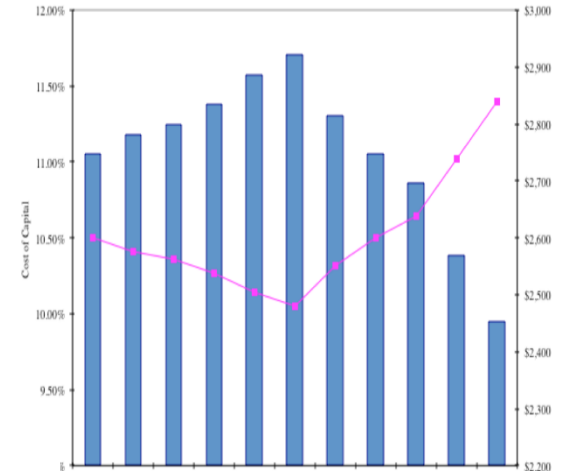
$$WACC = CMPC = r_e * (E/(D+E)) + (1-T)*r_d*(D/(D+E))$$

Optimal Capital Structure

Assume the firm has \$200 million in cash flows, expected to grow 3% a year forever.

D/(D+E)	Cost of Equity	After-tax Cost of Debt	Cost of Capital	Firm Value
0	10.50%	4.80%	10.50%	\$2,747
10%	11.00%	5.10%	10.41%	\$2,780
20%	11.60%	5.40%	10.36%	\$2,799
30%	12.30%	5.52%	10.27%	\$2,835
40%	13.10%	5.70%	10.14%	\$2,885
50%	14.50%	6.10%	10.30%	\$2,822
60%	15.00%	7.20%	10.32%	\$2,814
70%	16.10%	8.10%	10.50%	\$2,747
80%	17.20%	9.00%	10.64%	\$2,696
90%	18.40%	10.20%	11.02%	\$2,569
100%	19.70%	11.40%	11.40%	\$2,452

$$\text{Value} = \frac{\text{Expected Cash flow to firm next year}}{(\text{Cost of capital} - g)} = \frac{200(1.03)}{(\text{Cost of capital} - g)}$$



This is a simple example, where both the costs of debt and equity are given. The firm has cash flows, before debt and after taxes and reinvestment, of \$200 million and is in stable growth (growing 3% a year). Note that both increase as the debt ratio goes up, but the cost of capital becomes lower at least initially as you take on more debt (because you are substituting in cheaper debt for more expensive equity). At 40%, the cost of capital (WACC) is minimized. It is the optimal debt ratio.

To get firm value, we used the same cash flow (\$200 million) and same growth rate and changed the cost of capital.

Optimal Capital Structure

- Application exemple for company Disney
- The beta for Disney's stock in November 2013 was 1.0013. The T. bond rate at that time was 2.75%. Using an estimated equity risk premium of 5.76%, we estimated the cost of equity for Disney to be 8.52%:

$$\text{Cost of Equity} = 2.75\% + 1.0013(5.76\%) = 8.52\%$$

- Disney's bond rating in May 2009 was A, and based on this rating, the estimated pretax cost of debt for Disney is 3.75%. Using a marginal tax rate of 36.1, the after-tax cost of debt for Disney is 2.40%.

$$\text{After-Tax Cost of Debt} = 3.75\% (1 - 0.361) = 2.40\%$$

- The cost of capital was calculated using these costs and the weights based on market values of equity (121,878) and debt (15,961):

Cost of capital =

$$= 8.52\% \frac{121,878}{(15,961+121,878)} + 2.40\% \frac{15,961}{(15,961+121,878)} = 7.81\%$$

Optimal Capital Structure

- Steps to follow.... 1. Estimate the unlevered beta for the firm
- **The Regression Beta:** One approach is to use the regression beta (1.25) and then unlever, using the average debt to equity ratio (19.44%) during the period of the regression to arrive at an unlevered beta.

$$\text{Unlevered beta} = 1.25 / (1 + (1 - 0.361)(0.1944)) = 1.1119$$

- **The Bottom up Beta:** Alternatively, we can back to the source and estimate it from the betas of the businesses.

<i>Business</i>	<i>Revenues</i>	<i>EV/Sales</i>	<i>Value of Business</i>	<i>Proportion of Disney</i>	<i>Unlevered beta</i>	<i>Value</i>	<i>Proportion</i>
Media Networks	\$20,356	3.27	\$66,580	49.27%	1.03	\$66,579.81	49.27%
Parks & Resorts	\$14,087	3.24	\$45,683	33.81%	0.70	\$45,682.80	33.81%
Studio Entertainment	\$5,979	3.05	\$18,234	13.49%	1.10	\$18,234.27	13.49%
Consumer Products	\$3,555	0.83	\$2,952	2.18%	0.68	\$2,951.50	2.18%
Interactive	\$1,064	1.58	\$1,684	1.25%	1.22	\$1,683.72	1.25%
Disney Operations	\$45,041		\$135,132	100.00%	0.9239	\$135,132.11	100.00%

Optimal Capital Structure

	Most recent fiscal year (2012-13)	Prior year
Revenues	\$45,041	\$42,278
EBITDA	\$10,642	\$10,850
Depreciation & Amortization	\$2,192	\$1,987
EBIT	\$9,450	\$8,863
Interest Expenses	\$349	\$564
EBITDA (adjusted for leases)	\$12,517	\$11,168
Depreciation (adjusted for leases)	\$ 2,485	\$2,239
EBIT (adjusted for leases)	\$10,032	\$8,929
Interest Expenses (adjusted for leases)	\$459	\$630

Debt to Capital Ratio	D/E Ratio	Levered Beta	Cost of Equity
0%	0.00%	0.9239	8.07%
10%	11.11%	0.9895	8.45%
20%	25.00%	1.0715	8.92%
30%	42.86%	1.1770	9.53%
40%	66.67%	1.3175	10.34%
50%	100.00%	1.5143	11.48%
60%	150.00%	1.8095	13.18%
70%	233.33%	2.3016	16.01%
80%	400.00%	3.2856	21.68%
90%	900.00%	6.2376	38.69%

Levered Beta = $0.9239 (1 + (1 - .361) (D/E))$

Cost of equity = $2.75\% + \text{Levered beta} * 5.76\%$

Optimal Capital Structure

Start with the market value of the firm = $121,878 + \$15,961 = \$137,839$ million

D/(D+E) 0.00% 10.00% Debt to capital

D/E 0.00% 11.11% D/E = $10/90 = .1111$

\$ Debt \$0 \$13,784 10% of \$137,839

EBITDA \$12,517 \$12,517 Same as 0% debt

Depreciation \$ 2,485 \$2,485 Same as 0% debt

EBIT \$10,032 \$10,032 Same as 0% debt

Interest \$0 \$434 Pre-tax cost of debt * \$ Debt

Pre-tax Int. cov. ∞ 23.10 EBIT/ Interest Expenses

Likely Rating AAA AAA From Ratings table

Pre-tax cost of debt 3.15% 3.15% Riskless Rate + Spread

Optimal Capital Structure

<i>Interest coverage ratio is</i>	<i>Rating is</i>	<i>Spread is</i>	<i>Interest rate</i>
> 8.50	Aaa/AAA	0.40%	3.15%
6.5 – 8.5	Aa2/AA	0.70%	3.45%
5.5 – 6.5	A1/A+	0.85%	3.60%
4.25 – 5.5	A2/A	1.00%	3.75%
3 – 4.25	A3/A-	1.30%	4.05%
2.5 -3	Baa2/BBB	2.00%	4.75%
2.25 –2.5	Ba1/BB+	3.00%	5.75%
2 – 2.25	Ba2/BB	4.00%	6.75%
1.75 -2	B1/B+	5.50%	8.25%
1.5 – 1.75	B2/B	6.50%	9.25%
1.25 -1.5	B3/B-	7.25%	10.00%
0.8 -1.25	Caa/CCC	8.75%	11.50%
0.65 – 0.8	Ca2/CC	9.50%	12.25%
0.2 – 0.65	C2/C	10.50%	13.25%
<0.2	D2/D	12.00%	14.75%

T.Bond rate = 2.75%

Debt Ratio	\$ Debt	Interest Expense	Interest Coverage Ratio	Bond Rating	Pre-tax cost of debt	Tax rate	After-tax cost of debt
0%	\$0	\$0	∞	Aaa/AAA	3.15%	36.10%	2.01%
10%	\$13,784	\$434	23.10	Aaa/AAA	3.15%	36.10%	2.01%
20%	\$27,568	\$868	11.55	Aaa/AAA	3.15%	36.10%	2.01%
30%	\$41,352	\$1,427	7.03	Aa2/AA	3.45%	36.10%	2.20%
40%	\$55,136	\$2,068	4.85	A2/A	3.75%	36.10%	2.40%
50%	\$68,919	\$6,892	1.46	B3/B-	10.00%	36.10%	6.39%
60%	\$82,703	\$9,511	1.05	Caa/CCC	11.50%	36.10%	7.35%
70%	\$96,487	\$11,096	0.90	Caa/CCC	11.50%	32.64%	7.75%
80%	\$110,271	\$13,508	0.74	Ca2/CC	12.25%	26.81%	8.97%
90%	\$124,055	\$16,437	0.61	C2/C	13.25%	22.03%	10.33%

Optimal Capital Structure

- You need taxable income for interest to provide a tax savings. Note that the EBIT at Disney is \$10,032 million. As long as interest expenses are less than \$10,032 million, interest expenses remain fully tax-deductible and earn the 36.1% tax benefit. At an 60% debt ratio, the interest expenses are \$9,511 million and the tax benefit is therefore 36.1% of this amount.
- At a 70% debt ratio, however, the interest expenses balloon to \$11,096 million, which is greater than the EBIT of \$10,032 million. We consider the tax benefit on the interest expenses up to this amount:
 - Maximum Tax Benefit = EBIT * Marginal Tax Rate = \$10,032 million * 0.361 = \$3,622 million
 - Adjusted Marginal Tax Rate = Maximum Tax Benefit/Interest Expenses = \$3,622/\$11,096 = 32.64%

Debt Ratio	Beta	Cost of Equity	Cost of Debt (after-tax)	WACC
0%	0.9239	8.07%	2.01%	8.07%
10%	0.9895	8.45%	2.01%	7.81%
20%	1.0715	8.92%	2.01%	7.54%
30%	1.1770	9.53%	2.20%	7.33%
40%	1.3175	10.34%	2.40%	7.16%
50%	1.5143	11.48%	6.39%	8.93%
60%	1.8095	13.18%	7.35%	9.68%
70%	2.3762	16.44%	7.75%	10.35%
80%	3.6289	23.66%	8.97%	11.90%
90%	7.4074	45.43%	10.33%	13.84%

Optimal Capital Structure

Step 1: Estimate the cash flows to Disney as a firm

EBIT (1 – Tax Rate) = 10,032 (1 – 0.361) =	\$6,410
+ Depreciation and amortization =	\$2,485
– Capital expenditures =	\$5,239
– Change in noncash working capital	\$0
Free cash flow to the firm =	\$3,657

•Step 2: Back out the implied growth rate in the current market value

Current enterprise value = \$121,878 + 15,961 - 3,931 = 133,908

Value of firm = \$ 133,908

$$\begin{aligned}\text{Growth rate} &= (\text{Firm Value} * \text{Cost of Capital} - \text{CF to Firm}) / (\text{Firm Value} + \text{CF to Firm}) \\ &= (133,908 * 0.0781 - 3,657) / (133,908 + 3,657) = 0.0494 \text{ or } 4.94\%\end{aligned}$$

•Step 3: Revalue the firm with the new cost of capital

– Firm value = \$172,935

– Increase in firm value = \$172,935 - \$133,908 = \$39,027 million

Optimal Capital Structure

- In this approach, we start with the current market value and isolate the effect of changing the capital structure on the cash flow and the resulting value.

Enterprise Value before the change = \$133,908 million

Cost of financing Disney at existing debt ratio = \$ 133,908 * 0.0781 = \$10,458 million

Cost of financing Disney at optimal debt ratio = \$ 133,908 * 0.0716 = \$ 9,592 million

Annual savings in cost of financing = \$10,458 million – \$9,592 million = \$866 million

$$\text{Increase in Value} = \frac{\text{Annual Savings next year}}{(\text{Cost of Capital} - g)} = \frac{\$866}{(0.0716 - 0.0275)} = \$19,623 \text{ million}$$

Enterprise value after recapitalization

= Existing enterprise value + PV of Savings = \$133,908 + \$19,623 = \$153,531 million

- The reduction in the cost of capital translates into annual savings. Most of these savings are implicit, being savings in the cost of equity. Thus, the firm's accounting earnings will not reflect these savings directly.
- These savings can be converted into a present value by discounting back at the new cost of capital.
- It is more realistic to assume growth in firm value. A simple way to estimate what the current growth attributed to the firm by the market is to estimate it using the firm value today, the free cash flow to the firm and the current cost of capital.
- Note that the simple valuation formula used above assumes stable growth forever. For high growth firms, this formula will yield an implied growth rate that is too high (It will be very close to the cost of capital). In those cases, it is better to put a cap on the growth rate of around the risk free rate.
- In this case, maximizing firm value also maximizes stock price, because we assume that: Debt is refinanced at current market rates, thus protecting bondholders; Markets are rational and efficient.

Optimal Capital Structure

- Because the increase in value accrues entirely to stockholders, we can estimate the increase in value per share by dividing by the total number of shares outstanding (1,800 million).
 - Increase in Value per Share = $\$19,623 / 1800 = \$ 10.90$
 - New Stock Price = $\$67.71 + \$10.90 = \$78.61$
- Implicit in this computation is the assumption that the increase in firm value will be spread evenly across both the stockholders who sell their stock back to the firm and those who do not and that is why we term this the “rational” solution, since it leaves investors indifferent between selling back their shares and holding on to them
- If investors are rational, the price on the buyback should be such that investors are indifferent between selling back their shares and holding on to them. At \$78.61, that will happen.

Optimal Capital Structure

- Limitations of the use of the cost of capital to determine the optimal capital structure:
 - Static: The most critical number in the analysis is the operating result. If this changes, the optimal debt ratio will change.
 - Ignores indirect bankruptcy costs: Operating result is assumed as fixed, as the debt ratio, and the rating changes.
 - Beta and Ratings: it is based on strong assumptions of how market risk and the risk of bankruptcy are reborn as the company borrows more money and the resulting costs.

Optimal Capital Structure

- 2) For the reinforced capital cost:
- Costs of distress to affect operating results: In the cost of reinforced capital, the indirect costs of bankruptcy are included in expected operating results. As the company's rating decreases, the operating result is adjusted to reflect the RO loss that will occur when customers, suppliers and investors react.
- Dynamic analysis: Instead of looking at a single number in RO, we can obtain a distribution of RO's (allowing different results).

Optimal Capital Structure

<i>Rating</i>	<i>Drop in EBITDA (Low)</i>	<i>Drop in EBITDA (Medium)</i>	<i>Drop in EBITDA (High)</i>
To A	No effect	No effect	2.00%
To A-	No effect	2.00%	5.00%
To BBB	5.00%	10.00%	15.00%
To BB+	10.00%	20.00%	25.00%
To B-	15.00%	25.00%	30.00%
To C	25.00%	40.00%	50.00%
To D	30.00%	50.00%	100.00%

Debt Ratio	Beta	Cost of Equity	Bond Rating	Interest rate on debt	Tax Rate	Cost of Debt (after-tax)	WACC	Enterprise Value
0%	0.9239	8.07%	Aaa/AAA	3.15%	36.10%	2.01%	8.07%	\$122,633
10%	0.9895	8.45%	Aaa/AAA	3.15%	36.10%	2.01%	7.81%	\$134,020
20%	1.0715	8.92%	Aaa/AAA	3.15%	36.10%	2.01%	7.54%	\$147,739
30%	1.1769	9.53%	Aa2/AA	3.45%	36.10%	2.20%	7.33%	\$160,625
40%	1.3175	10.34%	A2/A	3.75%	36.10%	2.40%	7.16%	\$172,933
50%	1.5573	11.72%	C2/C	11.50%	31.44%	7.88%	9.80%	\$35,782
60%	1.9946	14.24%	Caa/CCC	13.25%	22.74%	10.24%	11.84%	\$25,219
70%	2.6594	18.07%	Caa/CCC	13.25%	19.49%	10.67%	12.89%	\$21,886
80%	3.9892	25.73%	Caa/CCC	13.25%	17.05%	10.99%	13.94%	\$19,331
90%	7.9783	48.72%	Caa/CCC	13.25%	15.16%	11.24%	14.99%	\$17,311

The optimal debt ratio stays at 40% but the cliff becomes much steeper.

Optimal Capital Structure

- Determinants of the optimal debt ratio:
- 1. The marginal tax rate
- The primary benefit of debt is the tax benefit. The higher the T , the greater the benefit associated with debt:

Tax Rate	Disney	Vale	Tata Motors	Baidu	Bookscape
0%	0%	0%	0%	0%	0%
10%	20%	0%	0%	0%	10%
20%	40%	0%	10%	10%	30%
30%	40%	30%	20%	10%	30%
40%	40%	40%	20%	10%	30%
50%	40%	40%	20%	10%	30%

Optimal Capital Structure

- 2. Return of cash flows prior to T
- Higher cash flows, such as % of value, will give greater indebtedness capacity, despite being lower in emerging markets with high country risk.

<i>Company</i>	<i>EBITDA</i>	<i>EBIT</i>	<i>Enterprise Value</i>	<i>EBITDA/EV</i>	<i>EBIT/EV</i>	<i>Optimal Debt</i>	<i>Optimal Debt Ratio</i>
Disney	\$12,517	\$10,032	\$133,908	9.35%	7.49%	\$55,136	40.00%
Vale	\$20,167	\$15,667	\$112,352	17.95%	13.94%	\$35,845	30.00%
Tata Motors	250,116₹	166,605₹	1,427,478₹	17.52%	11.67%	325,986₹	20.00%
Baidu	¥13,073	¥10,887	¥342,269	3.82%	3.18%	¥35,280	10.00%
Bookscape	\$4,150	\$2,536	\$42,636	9.73%	5.95%	\$13,091	30.00%

Optimal Capital Structure

- 3. Operational Risk
- Companies that face greater risk of uncertainty in their operations (and a higher variable operating result as a consequence) will have lower levels of optimal debt ratios than those with more predictable operations.
- Operational risks enter into the cost of capital in two ways:
- Non-Indebted Beta: Companies facing higher operational risk tend to have higher non-indebted betas. As they borrow, debt will further increase this high risk and tend to pull CP costs much more.
- Bond Ratings: For a given level of operating result, companies that face the greatest risk in operations will have lower ratings.

Optimal Capital Structure

- Continuation of the presentation of the ways to determine the OCS...
- 3. Present value adjusted to OCS (VALA)
- At adjusted net adjusted value, the value of the company is given by the value of this without debt plus the effect of the debt on the value of the company
- $\text{Company value} = \text{Value of non-indebted} + (\text{Tax benefits of debt} - \text{Expected value of debt bankruptcy costs})$
- The optimal value of debt will maximize the value of the company.
- How does VALA apply?

Optimal Capital Structure

- Step 1: Estimate the unlevered firm value. This can be done in one of two ways:
 - Estimating the unlevered beta, a cost of equity based upon the unlevered beta and valuing the firm using this cost of equity (which will also be the cost of capital, with an unlevered firm)
 - Alternatively, Unlevered Firm Value = Current Market Value of Firm - Tax Benefits of Debt (Current) + Expected Bankruptcy cost from Debt
- Step 2: Estimate the tax benefits at different levels of debt. The simplest assumption to make is that the savings are perpetual, in which case
 - Tax benefits = Dollar Debt * Tax Rate
- Step 3: Estimate a probability of bankruptcy at each debt level, and multiply by the cost of bankruptcy (including both direct and indirect costs) to estimate the expected bankruptcy cost.

Optimal Capital Structure

- Probability of Bankruptcy
 - Estimate the synthetic rating that the firm will have at each level of debt
 - Estimate the probability that the firm will go bankrupt over time, at that level of debt (Use studies that have estimated the empirical probabilities of this occurring over time - Altman does an update every year)
- Cost of Bankruptcy
 - The direct bankruptcy cost is the easier component. It is generally between 5-10% of firm value, based upon empirical studies
 - The indirect bankruptcy cost is much tougher. It should be higher for sectors where operating income is affected significantly by default risk (like airlines) and lower for sectors where it is not (like groceries)

Optimal Capital Structure

Rating	Likelihood of Default
AAA	0.07%
AA	0.51%
A+	0.60%
A	0.66%
A-	2.50%
BBB	7.54%
BB	16.63%
B+	25.00%
B	36.80%
B-	45.00%
CCC	59.01%
CC	70.00%
C	85.00%
D	100.00%

Altman estimated these probabilities by looking at bonds in each ratings class ten years prior and then examining the proportion of these bonds that defaulted over the ten years.

Optimal Capital Structure

- Example Disney

Current Value of firm = \$121,878 + \$15,961 = \$ 137,839

- Tax Benefit on Current Debt = \$15,961 * 0.361 = \$ 5,762

+ Expected Bankruptcy Cost = 0.66% * (0.25 * 137,839) = \$ 227

Unlevered Value of Firm = = \$ 132,304

- Cost of Bankruptcy for Disney = 25% of firm value
- Probability of Bankruptcy = 0.66%, based on firm's current rating of A
- Tax Rate = 36.1%

Optimal Capital Structure

<i>Debt Ratio</i>	<i>\$ Debt</i>	<i>Tax Rate</i>	<i>Unlevered Firm Value</i>	<i>Tax Benefits</i>	<i>Bond Rating</i>	<i>Probability of Default</i>	<i>Expected Bankruptcy Cost</i>	<i>Value of Levered Firm</i>
0%	\$0	36.10%	\$132,304	\$0	AAA	0.07%	\$23	\$132,281
10%	\$13,784	36.10%	\$132,304	\$4,976	Aaa/AAA	0.07%	\$24	\$137,256
20%	\$27,568	36.10%	\$132,304	\$9,952	Aaa/AAA	0.07%	\$25	\$142,231
30%	\$41,352	36.10%	\$132,304	\$14,928	Aa2/AA	0.51%	\$188	\$147,045
40%	\$55,136	36.10%	\$132,304	\$19,904	A2/A	0.66%	\$251	\$151,957
50%	\$68,919	36.10%	\$132,304	\$24,880	B3/B-	45.00%	\$17,683	\$139,501
60%	\$82,703	36.10%	\$132,304	\$29,856	C2/C	59.01%	\$23,923	\$138,238
70%	\$96,487	32.64%	\$132,304	\$31,491	C2/C	59.01%	\$24,164	\$139,631
80%	\$110,271	26.81%	\$132,304	\$29,563	Ca2/CC	70.00%	\$28,327	\$133,540
90%	\$124,055	22.03%	\$132,304	\$27,332	Caa/CCC	85.00%	\$33,923	\$125,713

The optimal debt ratio is 40%, which is the point at which firm value is maximized.

Optimal Capital Structure

4) Relative analysis (Continuation of presentation of ways to determine OCS ...):

The safest place for any business is to be close to the industry average

Subjective adjustments can be made to these averages to arrive at the correct debt ratio.

Higher tax rates -> Higher debt ratios (Higher tax benefits)

Lower internal ownership -> Higher debt ratios (Higher discipline)

Stable income -> Higher debt ratios (Lower bankruptcy costs)

More intangible assets -> lower debt ratios (Larger agency problems)

Optimal Capital Structure

	<i>Debt to Capital Ratio</i>		<i>Net Debt to Capital Ratio</i>			<i>Debt to Capital Ratio</i>		<i>Net Debt to Capital Ratio</i>	
<i>Company</i>	<i>Book value</i>	<i>Market value</i>	<i>Book value</i>	<i>Market value</i>	<i>Comparable group</i>	<i>Book value</i>	<i>Market value</i>	<i>Book value</i>	<i>Market value</i>
Disney	22.88%	11.58%	17.70%	8.98%	US Entertainment	39.03%	15.44%	24.92%	9.93%
Vale	39.02%	35.48%	34.90%	31.38%	Global Diversified Mining & Iron Ore (Market cap > \$1 b)	34.43%	26.03%	26.01%	17.90%
Tata Motors	58.51%	29.28%	22.44%	19.25%	Global Autos (Market Cap > \$1 b)	35.96%	18.72%	3.53%	0.17%
Baidu	32.93%	5.23%	20.12%	2.32%	Global Online Advertising	6.37%	1.83%	-27.13%	-2.76%

Optimal Capital Structure

Get past simple averages

Step 1: Run a regression of debt ratios on the variables that you believe determine debt ratios in the sector. For example,

$$\text{Debt Ratio} = a + b (\text{Tax rate}) + c (\text{Earnings Variability}) + d (\text{EBITDA/Firm Value})$$

Check this regression for statistical significance (t statistics) and predictive ability (R squared)

Step 2: Estimate the values of the proxies for the firm under consideration. Plugging into the cross sectional regression, we can obtain an estimate of predicted debt ratio.

Step 3: Compare the actual debt ratio to the predicted debt ratio.

Additional notes about dividend policy

Appendix

Looking at the market!

- Doing a A regression with dividend yield and payout data against expected growth in US companies in January 2014 yielded::

$$\text{PYT} = 0.649 - 0.296 (\text{BETA}) - .800 (\text{EGR}) + .300 (\text{DCAP}) \quad R^2 = 19.6\%$$

(32.16) (15.40) (8.90) (7.33)

$$\text{YLD} = 0.0324 - .0154 (\text{BETA}) - .038 (\text{EGR}) + .023 (\text{DCAP}) \quad R^2 = 25.8\%$$

(38.81) (19.41) (13.25) (13.45)

PYT = Dividend Payout Ratio = Dividends/Net Income

YLD = Dividend Yield = Dividends/Current Price

BETA = Beta (Regression or Bottom up) for company

EGR = Expected growth rate in earnings over next 5 years (analyst estimates)

DCAP = Total Debt / (Total Debt + Market Value of equity)

Using Disney regression

- To illustrate the applicability of the regression in Disney's dividend policy analysis, we estimated the values of the independent variables in the company regressions.
- Beta for Disney (bottom up) = 1.00
- Disney's expected growth in earnings per share = 14.73% (analyst estimate)
- Disney's market debt to equity ratio = 11.58%
- Substituting in the regression equations for the dividend payout ratio and the dividend yield, a predictable payout ratio was estimated:
- Predicted Payout = $.649 - 0.296 (1.00) - .800 (.1473) + .300 (.1158) = .2695$
- Predicted Yield = $0.0324 - .0154 (1.00) - .38 (.1473) + .023 (.1158) = .0140$
- Based on this analysis, Disney with its dividend yield of 1.09% and a payout ratio of approximately 21.58% is paying very little of dividends. This analysis, however, fails because it does not consider the high stock buybacks made by Disney in recent years.

Dividends Policy

Dividends

They represent the basic return variable with which owners and investors determine the value of the share.

They are a source of cash flow to shareholders and provide information on the current and future performance of the company.

Retained earnings, ie profits not distributed as dividends to shareholders, represent a form of internal financing, in this sense, the decision of dividends may strongly affect the requirements of external financing (the higher the amount of the dividend paid, the higher the the volume of funding to be obtained externally).

Dividends Policy

- The term dividend refers to the money distributed to the shareholders on account of the results generated by the company.
- Generally, any direct payment made by the company to the shareholders can be considered as a dividend or part of the dividend policy.
- Execution of the dividend policy => decisions of distribution of results.
- Do these decisions influence the value of the company?
- Will they be decisive in the financial function of the company?

Dividends Policy

Dividends

Dividends tend to accompany the phenomenon of increased profits (increases profits, increases the distribution of dividends);

However, sometimes this increase in dividends does not keep up with the proportion of the increase in profits, which means that the company retains profits to add value to the company to invest in profitable projects.

Another situation may occur that passes through the distribution of dividends through stock Buy-back.

Stock (Share) Buy-back (repurchase of own shares)

Currently many managers are paid through the company's share repurchase, with the purpose of valuing the company's assets, since the higher the value of the company's assets, the greater the value of the shares;

Another form of shareholder remuneration;

Tax advantage - less taxation of capital gains on dividends

Since the 1990s, this form of dividend payment has become more widely used than the distribution of dividends;

Dividends Policy

Different Types of Dividends

1) Payment in shares (free allocation of shares): These may be common or preferred shares; The rule is to maintain the shareholder structure, that is, the payment is made in proportion to the capital held so as not to change the relative position of each shareholder; It leads to an increase in the number of shares; Incorporation of reserves or stock split! It is a purely fiscal advantage since there is less taxation of capital gains in relation to dividends.

Stock split: In an operation of this type there is no cash inflow by investors; It is only an accounting operation of dividing the par value of the shares. The company has the same capital only divided by a greater number of shares. Example: In Portugal most companies divided the par value of shares from \$ 1,000 to \$ 1, so each investor holding a share now holds about 5 shares.

Dividends Policy

Different Types of Dividends

- 2) Cash Payment: More generalized method of payment of dividends;
- 3) Repurchase of shares: Appropriate modality for companies with high liquidity; It conveys to the market the "idea" that stocks are undervalued. With this operation the number of shares decreases but the price per share rises; It has the same effect as a cash dividend: there is a cash outflow from the company in favor of the shareholders.
- 4) Payment in kind (cars, watches, houses, trips, etc.).

Dividends Policy

- Article 294 of the Commercial Companies Code - half of the distributable profit must be delivered to the shareholders; Unless the vote is unanimous (a majority of 2/3), to the contrary.
- Article 71 of the IRS Code - Withholding tax definitively, at the liberation rate of 28% (as of 12/31/2012) of the amount of dividends distributed.
- Legal limits on the distribution of dividends:
- Portuguese legislation has as its general principle an annual payment of dividends.
- However it allows companies to make interim dividend payments in the form of "advanced", but only once a year.
- Portuguese companies may not distribute dividends when:
- There are negative results
- Legal reserves are not yet formed
- Formation, research and development expenses are not amortized, unless there are free reserves in the same amount.
- In the US, the rule is a quarterly payment of dividends - bringing the momentum of the yield generated by the shares from the moment of the income generated by the bonds.

Dividends Policy

Dividend payment procedure

The payment of dividends or not in cash is decided by the members of the board of directors, in quarterly or semiannual meetings, a policy that is accepted by the shareholders.

The "recent" past financial performance, as well as the prospects for future performance, are basic elements for the dividend decision.

The payment date, if it is distributed in cash, should also be decided in a meeting.

Dividends Policy

Dividend payment procedure (cont.)

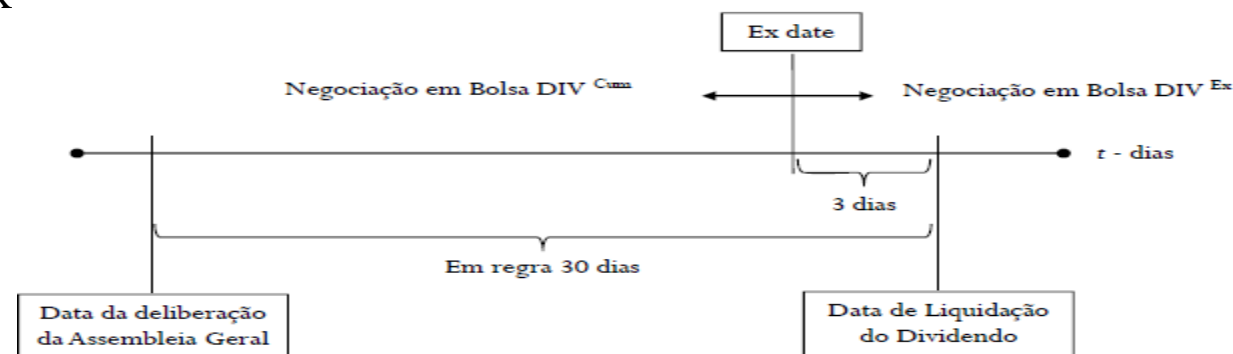
Dividend volume

The dividend policy determines whether or not to be paid and how much.

Most companies have an established periodic dividend policy, however, directors may change this amount, primarily based on highs or significant decreases in profits.

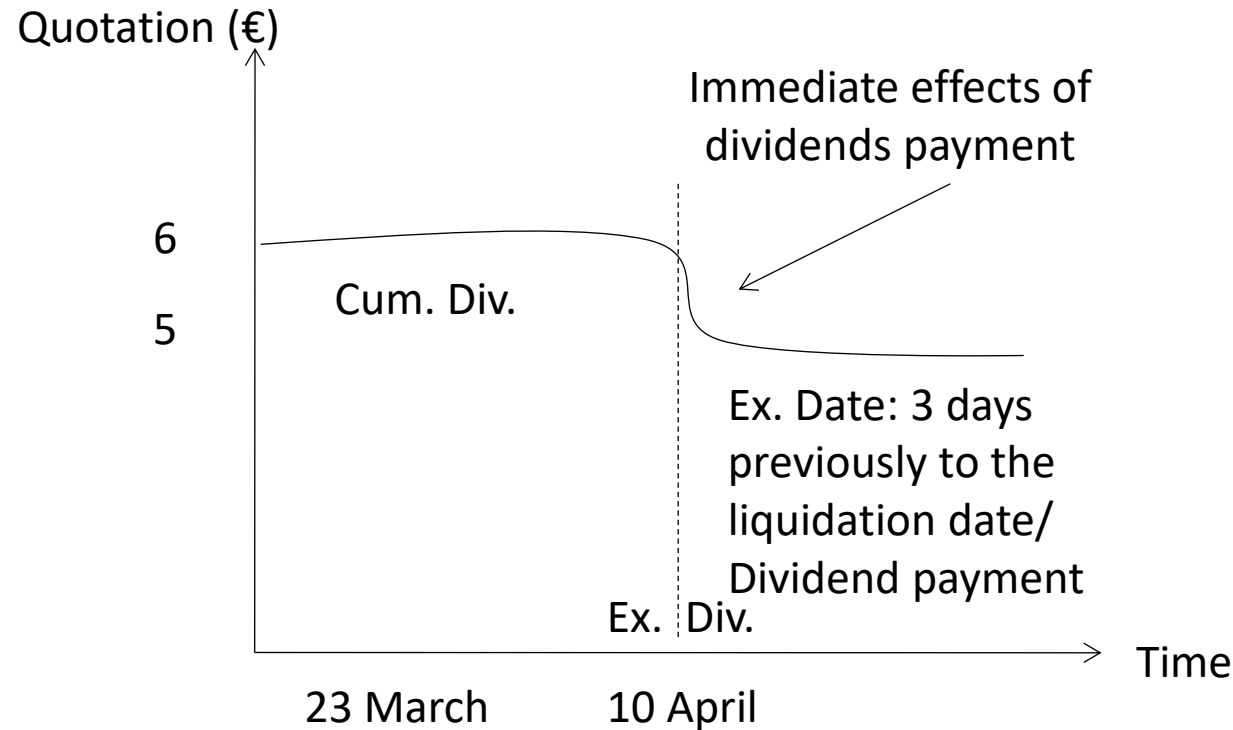
Phases of dividends distribution

- Phase 1 - Deliberation of the General Shareholders' Meeting with the proposal of distribution of results. Including the dividend amount to be distributed.
- Phase 2 - Date of payment of the dividend to shareholders. (as a rule, up to 30 days after the resolution of the AG.)
- Between these two dates there is an important corporate event:
- Trading of Cum DIV shares:
- Trading on the stock exchange entitled to the dividend. Up to three days before the dividend payment date.
- Trading of Ex Div Shares:
- Trading on the stock exchange without dividend rights. Three days before the dividend payment date.



Quotation effect

- What is the effect of the share price of a dividend payment:



When dividends are paid, theoretically the price should be reduced by the same amount as the distributed dividend. However, in practice, such an adjustment does not always take place in full, which makes it possible to verify that the price of securities increased.

Dividends Policy

- Companies opt for stable dividend distribution policies.
- Studies have shown that markets react negatively to the distribution of dividends:
- => If dividends are a function of Net Results and there are dividend increases, there is less money to cope with the future growth of the company (which explains the negative reaction of the market)
- On the other hand, the market can also react positively to this announcement if the company's Net Results also grow.
- Problem: Will it remain so always high? Or will it go downhill from there?

Dividends Policy

Factors Affecting Dividend Policy

The dividend policy must be consistent with the company's overall objectives.

Some factors are considered in the definition of these policies:

- Legal restrictions;
- Contractual restrictions;
- Internal restrictions;
- Growth prospects;
- Considerations relating to owners;
- Market considerations (within these we have the reactions to the announcements of increase or decrease of dividends to distribute - sensitive to listed companies).

Dividends Policy

Factors Affecting Dividend Policy (cont.)

1) Legal Restrictions

Most countries prohibit the full distribution of dividends in order to restrict capital reduction so as to ensure that there is sufficient capital to protect the rights of creditors.

2) Contractual Restrictions

The ability to pay dividends of a company is limited by restrictive clauses, which usually prohibit the payment of cash dividends up to a certain percentage of profits or up to a certain level of monetary value. These restrictions help protect creditors against losses arising from the insolvency of the company.

Dividends Policy

Factors Affecting Dividend Policy (cont.)

3) Internal Restrictions

The ability to pay cash dividends is limited by the volume of net assets (cash and marketable securities) available.

While it is possible to use the value of loans to pay dividends, fund providers typically "condemn" the realization of these loans, since they do not produce tangible or operational benefits for the company.

Dividends Policy

Factors Affecting Dividend Policy (cont.)

4) Growth Perspectives

The financial needs of the company are directly related to its expectation of growth and the assets that it will need to acquire.

In this sense, the company must evaluate its profitability and its risk, in order to evaluate its ability to obtain external resources.

In addition, you need to determine the cost and speed at which you can get funding.

All this influences the decision of the dividend policy.

Dividends Policy

Factors Affecting Dividend Policy (cont.)

4) Growth Prospects (cont.)

Large companies in the Maturity phase - usually have adequate access to new capital;

Fast-growing businesses - may not have enough funds to finance their projects;

Growing companies - may rely heavily on domestic financing through retained earnings, which leads to a dividend policy based on a small proportion of profits.

A more consolidated company - it will be in a better position to distribute a high proportion of profits, especially if funding sources are readily available.

Dividends Policy

Factors Affecting Dividend Policy (cont.)

5) Owner Considerations

The company must set a policy that will have a positive effect on the wealth of most owners.

One important consideration is the tax status of homeowners. If there is a high percentage of wealthy shareholders with a high tax rate, a decision can be made to pay a lower proportion of profits to allow owners to delay paying taxes until they sell the shares.

Important Note - It is illegal to determine the dividend policy by the tax status of the owners. Although they are recurrent.

Dividends Policy

Factors Affecting Dividend Policy (cont.)

5) Owner Considerations (cont.)

Another important issue are the investment opportunities available to homeowners.

You should only opt for internal retention when your company has projects that deliver the highest return.

Another situation that may occur is the potential dilution of property.

If a company distributes a high proportion of profits, it will be necessary to raise the new equity with the issuance of common shares.

The result of the issuance of shares may be the dilution of control and profits of existing shareholders.

By distributing a smaller proportion of profits to the company you can minimize the possibility of such dilution.

Dividends Policy

Factors Affecting Dividend Policy (cont.)

6) Market Considerations

When formulating the dividend policy it is important to take into account the likely reaction of the markets.

Shareholders prefer a fixed or increasing level of dividend policy compared to a floating level policy.

They also prefer an uninterrupted payment of dividends; A regular or increasing payment eliminates the uncertainty as to the frequency of the magnitude of the dividends. In this way, the company's returns tend to be discounted at a lower rate, which results in an increase in the market value of the shares and then an expansion of the owners' wealth.

Dividends Policy

Factors Affecting Dividend Policy (cont.)

6) Market Considerations (cont.)

Failure to pay dividends may generate uncertainty about the company's future and may lead to a decrease in the value of the share.

NOTES

Companies should never refuse a project with a positive NPV (Net Present Value) instead of paying dividends, since a project is an opportunity to create value for the company.

If a project increases the value of the company, its equity, many shareholders prefer not to receive dividends, and may have advantages in the future, as the company is increasing the value of its assets.

Determinants of the dividend policy

Investment opportunities: if there are investment opportunities (positive VAL's) we will distribute less dividends (or none at all);

VAL of the company's cash flows = Receipts - Payments = -
Investment + RL + Depreciation + Provisions / Impairments +
Financial Financing Charges

Stability of profits: greater profit stability conveys a positive image to the market; there is a greater tendency to pay more dividends (solid company);

The degree of access to other financing alternatives, particularly the banking institutions, influences the distribution of dividends;

Restrictions on bondholders and banks may restrict the distribution of dividends;

The degree of risk aversion of shareholders / shareholders influences the dividend policy